Data Mining Project

Health Insurance Coverage and Hypertension Risk Analysis

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Content

Problem Statement

→

Insurance Disparities

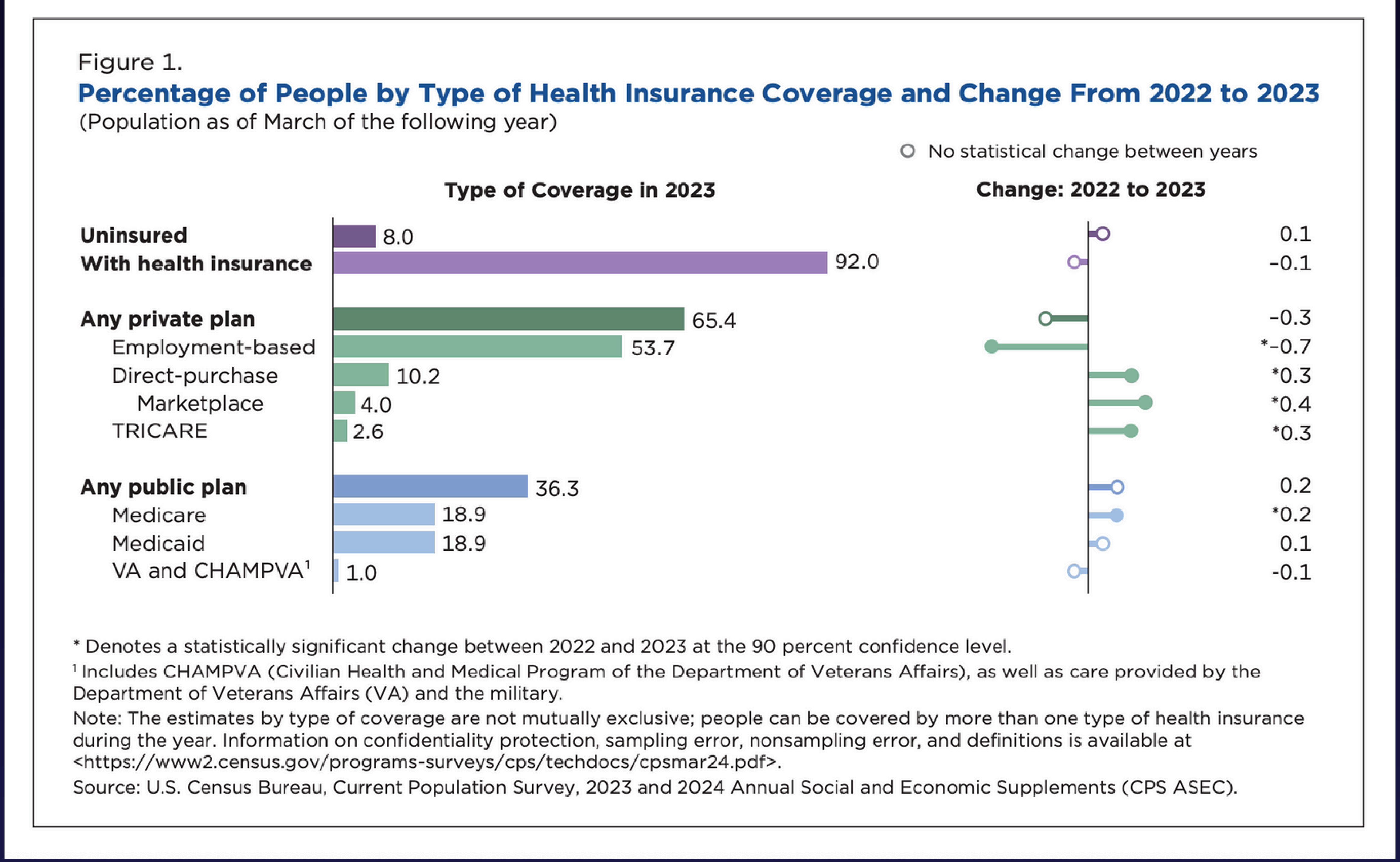
Hypertension Burden

Insurance disparities

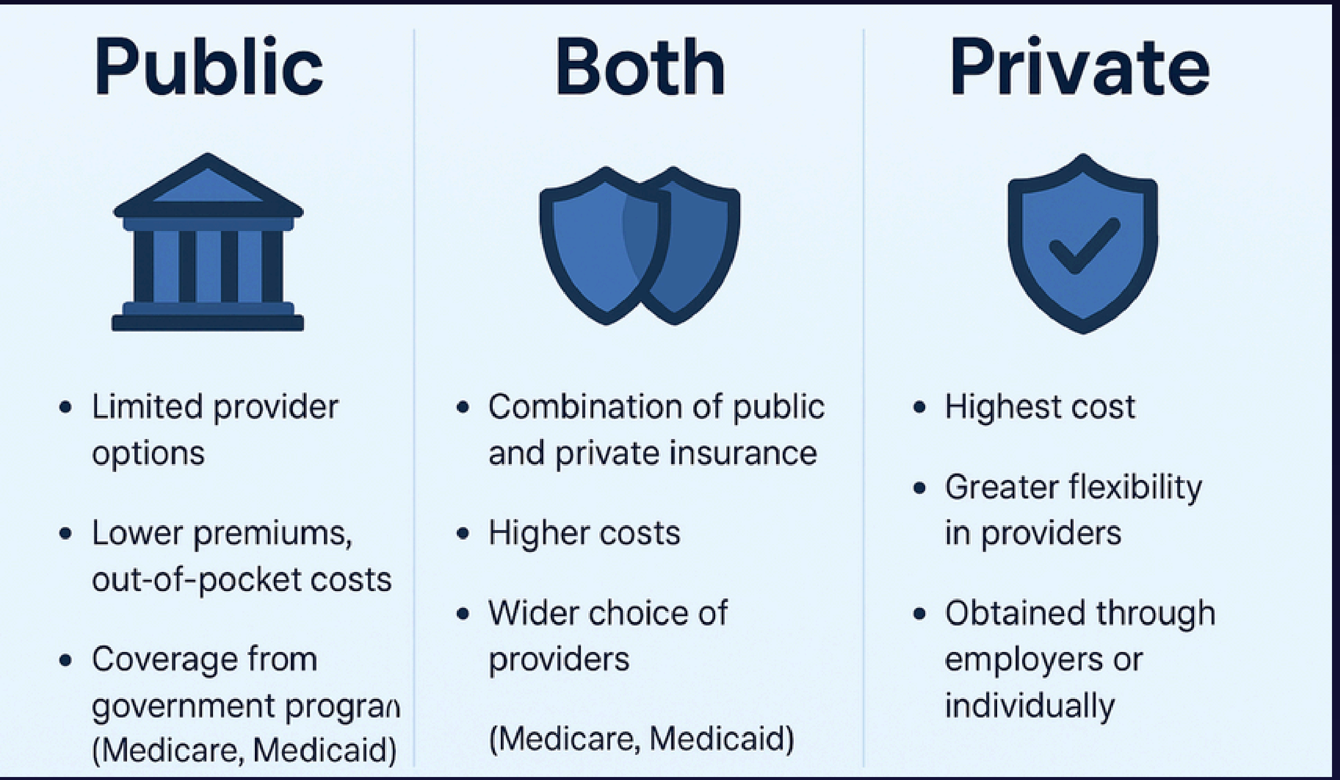
→

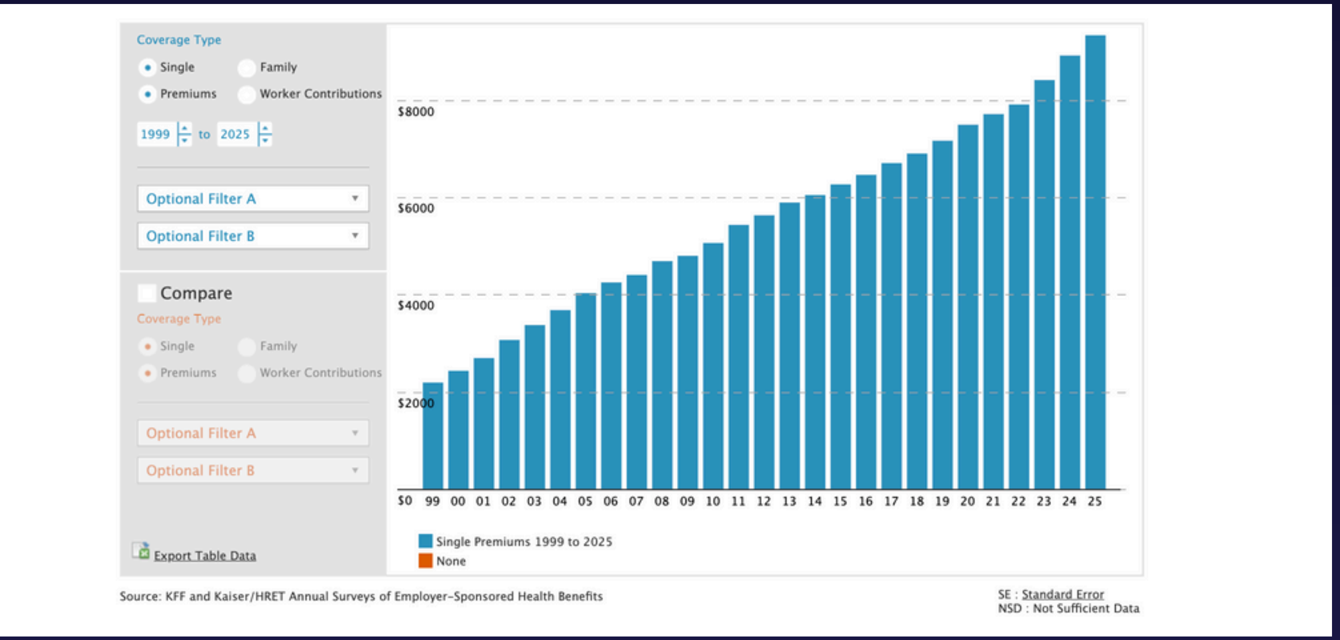
92% **Insurance Coverage**

But Insurance Inequities Persist



Insurance Coverage ≠ Equal Access



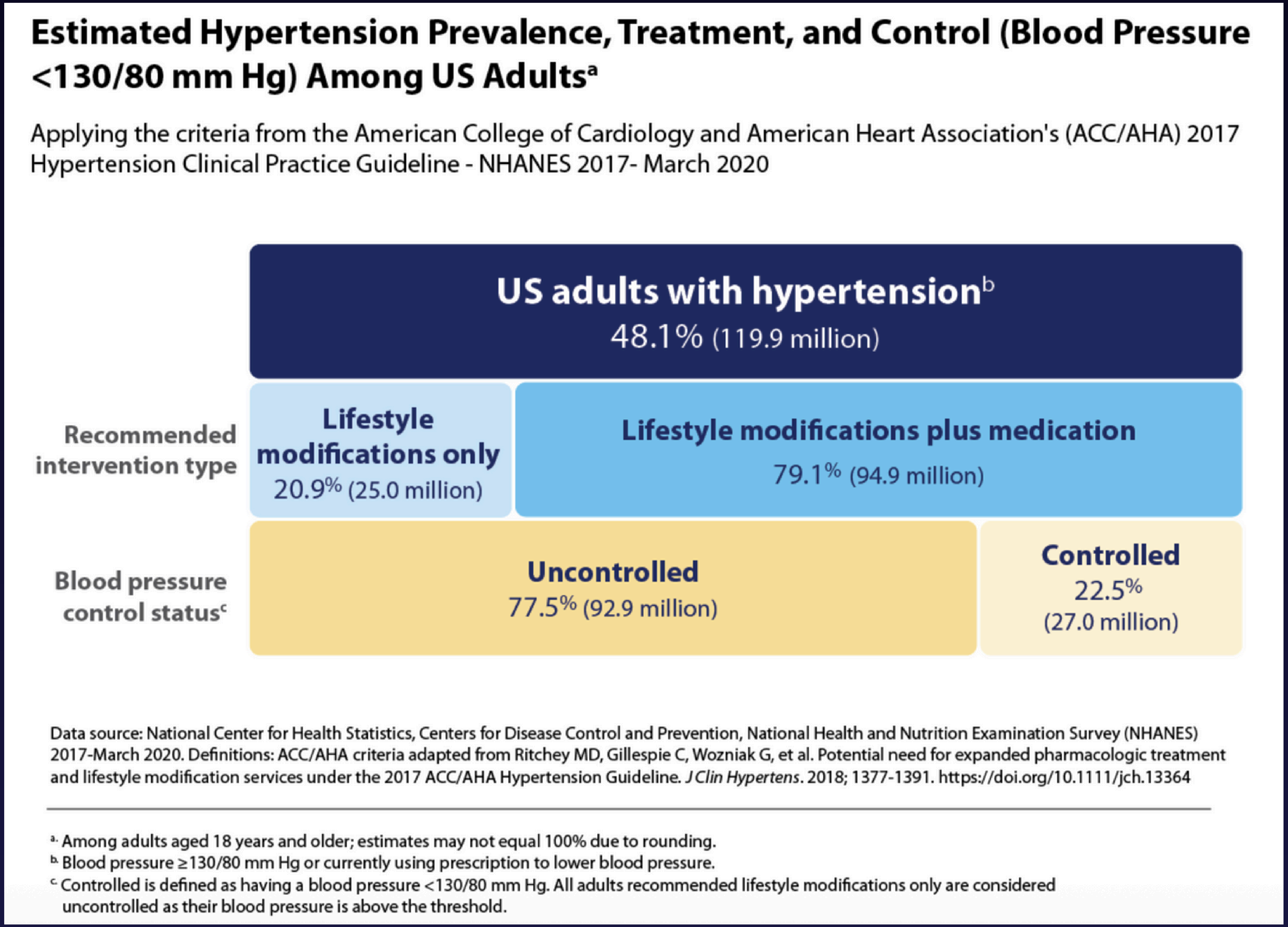


Hypertension Burden

48.1%

Hypertension Prevalence

Only **25%** manage to control



[1] https://www.cdc.gov/high-blood-pressure/data-research/facts-stats/index.html

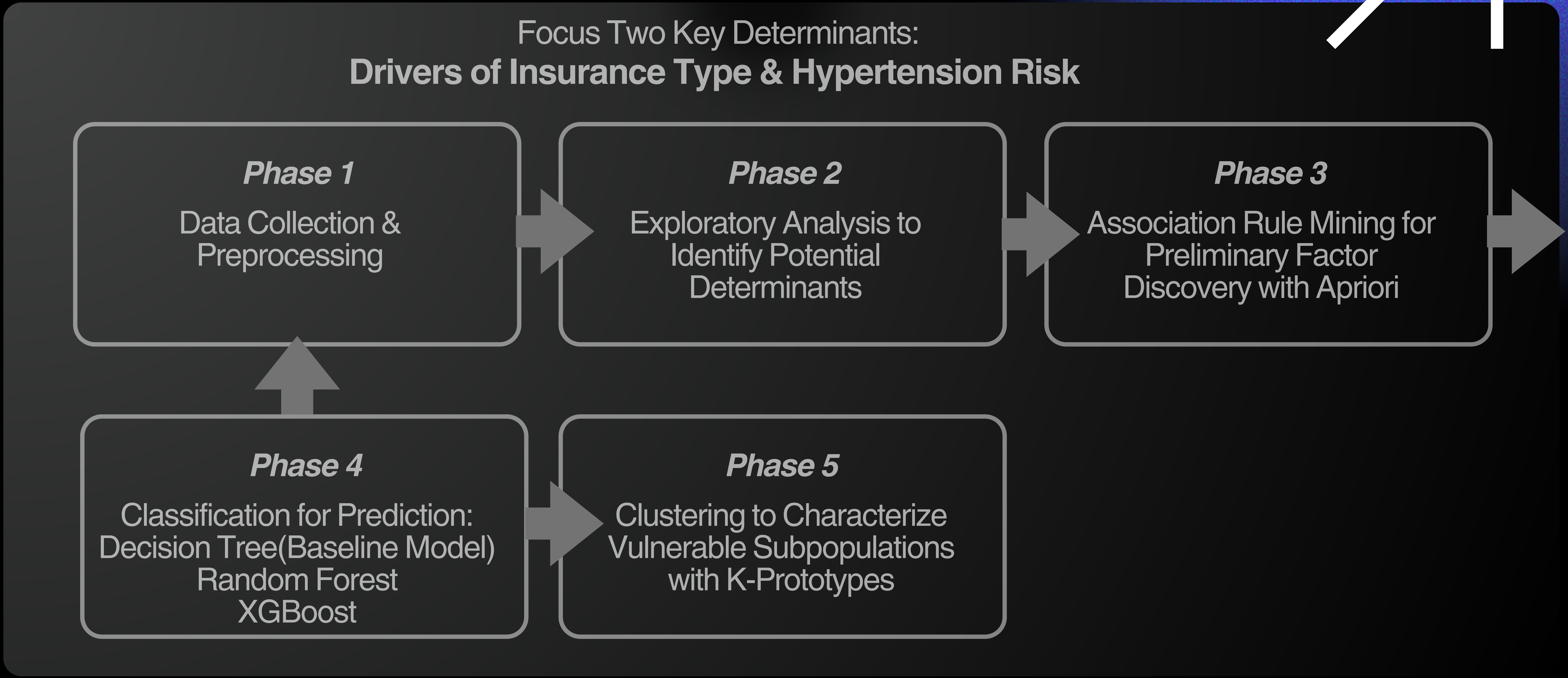
What Factors Shape Insurance Choice and Hypertension Risk?

Identify
Key Patterns
Behind Insurance
Types

Predict
Hypertension
Risk

Detect vulnerable population subgroups

Project Workflow



Centers for Disease Control and Prevention
CDC 24/7: Saving Lives, Protecting People™

National Health and Nutrition Examination Survey

Dataset Description

NHANES August 2021-August 2023

11,933 observations x **154** variables

Data, Documentation, Codebooks

- Demographics Data
- Dietary Data
- Examination Data
- Laboratory Data
- Questionnaire Data
- Limited Access Data

Data File Name	Doc File	Data File	Date Published
Audiometry	AUQ_L Doc	AUQ_L Data [XPT - 1.3 MB]	September 2024
Acculturation	ACQ_L Doc	ACQ_L Data [XPT - 445.6 KB]	September 2024
Alcohol Use	ALQ_L Doc	ALQ_L Data [XPT - 447.6 KB]	September 2024
Balance	BAQ_L Doc	BAQ_L Data [XPT - 714.1 KB]	October 2024
Blood Pressure & Cholesterol	BPQ_L Doc	BPQ_L Data [XPT - 400.1 KB]	September 2024
Current Health Status	HSQ_L Doc	HSQ_L Data [XPT - 104.4 KB]	September 2024
Dermatology	DEQ_L Doc	DEQ_L Data [XPT - 135.8 KB]	September 2024
Diabetes	DIQ_L Doc	DIQ_L Data [XPT - 827.7 KB]	September 2024
Diet Behavior & Nutrition	DBQ_L Doc	DBQ_L Data [XPT - 2.5 MB]	September 2024

```
       SEQN  SDDSRVYR  RIDSTATR  RIAGENDR  RIDAGEYR  RIDAGEMN  RIDRETH1  \
0  130378.0      12.0       2.0       1.0      43.0       NaN       5.0
1  130379.0      12.0       2.0       1.0      66.0       NaN       3.0
2  130380.0      12.0       2.0       2.0      44.0       NaN       2.0
3  130381.0      12.0       2.0       2.0       5.0       NaN       5.0
4  130382.0      12.0       2.0       1.0       2.0       NaN       3.0

   RIDRETH3  RIDEXMON  RIDEXAGM  ...  ALQ111  ALQ121  ALQ130        ALQ142  \
0       6.0       2.0       NaN  ...     NaN     NaN     NaN           NaN
1       3.0       2.0       NaN  ...     1.0     2.0     3.0  5.397605e-79
2       2.0       1.0       NaN  ...     1.0    10.0     1.0  5.397605e-79
3       7.0       1.0      71.0  ...     NaN     NaN     NaN           NaN
4       3.0       2.0      34.0  ...     NaN     NaN     NaN           NaN

   ALQ270  ALQ280  ALQ151  ALQ170     WTPH2YR_y  LBXAGP
0     NaN     NaN     NaN     NaN           NaN     NaN
1     NaN     NaN     2.0     NaN           NaN     NaN
2     NaN     NaN     2.0     NaN  8.532884e+04   1.010
3     NaN     NaN     NaN     NaN  5.397605e-79     NaN
4     NaN     NaN     NaN     NaN  5.963893e+04   0.921

[5 rows x 154 columns]
```

– Demographics Data (DEMO: age, gender, race, education, income-to-poverty ratio)
– Examination Data (BMX, BPXO: physical measurements such as BMI, sbp, dbp)
– Laboratory Data (AGP, GLU: clinical indicators (e.g., a1c, glucose))
– Questionnaire Data (DIQ, HIQ, ect.: self-reported lifestyle behaviors, health conditions, insurance coverage (e.g., smoking, physical activity, sleep hours, diabetes, insurance type))

Select **17 individual NHANES survey component files** & Merge by a unique sequence number (SEQN)

https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx

Dataset Preprocessing

```
       SEQN  SDDSRVYR  RIDSTATR  RIAGENDR  RIDAGEYR  RIDAGEMN  RIDRETH1  \
0  130378.0      12.0       2.0       1.0      43.0       NaN       5.0
1  130379.0      12.0       2.0       1.0      66.0       NaN       3.0
2  130380.0      12.0       2.0       2.0      44.0       NaN       2.0
3  130381.0      12.0       2.0       2.0       5.0       NaN       5.0
4  130382.0      12.0       2.0       1.0       2.0       NaN       3.0

   RIDRETH3  RIDEXMON  RIDEXAGM  ...  ALQ111  ALQ121  ALQ130        ALQ142  \
0       6.0       2.0       NaN  ...     NaN     NaN     NaN           NaN
1       3.0       2.0       NaN  ...     1.0     2.0     3.0  5.397605e-79
2       2.0       1.0       NaN  ...     1.0    10.0     1.0  5.397605e-79
3       7.0       1.0      71.0  ...     NaN     NaN     NaN           NaN
4       3.0       2.0      34.0  ...     NaN     NaN     NaN           NaN

   ALQ270  ALQ280  ALQ151  ALQ170     WTPH2YR_y  LBXAGP
0     NaN     NaN     NaN     NaN           NaN     NaN
1     NaN     NaN     2.0     NaN           NaN     NaN
2     NaN     NaN     2.0     NaN  8.532884e+04   1.010
3     NaN     NaN     NaN     NaN  5.397605e-79     NaN
4     NaN     NaN     NaN     NaN  5.963893e+04   0.921

[5 rows x 154 columns]
```

11,933 observations x **154** variables

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Diabetes (DIQ_L)

Data File: DIQ_L.xpt

DIQ010 - Doctor told you have diabetes

Variable Name: DIQ010

SAS Label: Doctor told you have diabetes

English Text: The next questions are about specific medical conditions. {Other than during pregnancy, {have you/has SP}/{Have you/Has SP}} ever been told by a doctor or health professional that {you have/{he/she/SP} has} diabetes or sugar diabetes?

English Instructions: CAPI INSTRUCTION: IF SP AGE < 15, DISPLAY "HAVE SP" FOR THE FIRST DISPLAY AND "SP HAS" FOR THE SECOND DISPLAY. IF SP IS FEMALE AND AGE >= 20, DISPLAY "OTHER THAN DURING PREGNANCY, {HAVE YOU/HAS SP}".

Target: Both males and females 1 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	1081	1081	
2	No	10371	11452	DIQ159
3	Borderline	284	11736	DIQ159
7	Refused	0	11736	DIQ159
9	Don't know	4	11740	DIQ159
.	Missing	4	11744	

Variable renaming and distinct value decode based on the ***Codebook***

	id	age	gender	race	education	income_poverty_ratio	weight_kg	height_cm	bmi	sbp	...	a1c	glucose
0	130378.0	43.0	Male	Non-Hispanic Asian	College graduate or above	5.00	86.9	179.5	27.0	135.0	...	5.6	113.0
1	130379.0	66.0	Male	Non-Hispanic White	College graduate or above	5.00	101.8	174.2	33.5	121.0	...	5.6	99.0
2	130380.0	44.0	Female	Other Hispanic	High school/GED	1.41	69.4	152.9	29.7	111.0	...	6.2	156.0
3	130381.0	5.0	Female	Other Race / Multi	NaN	1.53	34.3	120.1	23.8	NaN	...	NaN	NaN
4	130382.0	2.0	Male	Non-Hispanic White	NaN	3.60	13.6	NaN	NaN	NaN	...	NaN	NaN

– Demographics: age, gender, education, income poverty ratio
– Health indicators: sbp, dbp, bmi, diabetes self report
– Lifestyle:smoking current, smoking ever, physical activity freq, sleep hours
– Insurance status: insurance type

11,933 observations x **21** variables

Data Preprocessing

	id	age	income_poverty_ratio	weight_kg	height_cm	bmi	sbp	dbp	a1c	glucose	physical_a
count	11933.00	11933.00	9892.00	8754.00	8499.00	8471.00	7517.00	7517.00	6715.00	3672.00	
mean	136344.00	38.32	2.71	70.55	159.66	27.25	119.29	72.75	5.71	107.88	
std	3444.90	25.60	1.67	30.39	19.86	8.14	18.56	11.90	1.05	32.48	
min	130378.00	0.00	0.00	2.70	79.10	11.10	61.00	33.00	3.20	59.00	
25%	133361.00	13.00	1.18	54.20	154.40	21.60	106.00	64.00	5.20	93.00	
50%	136344.00	37.00	2.50	71.70	163.60	26.40	117.00	72.00	5.50	100.00	
75%	139327.00	62.00	4.50	89.10	172.10	31.70	130.00	80.00	5.80	109.00	
max	142310.00	80.00	5.00	248.20	200.70	74.80	232.00	142.00	17.10	561.00	

	gender	race	education	smoking_current	smoking_ever	diabetes_self_report	has_health_insurance	doctor_visit_12
count	11933	11933	7783	8135	1190	11740	11910	67
unique	2	6	5	4	2	4	4	
top	Female	Non-Hispanic White	College graduate or above	No	Yes	No	Insured	[illegible]
freq	6358	6217	2625	4878	952	10371	11007	36[illegible]

Data Summary for Outlier Check

	missing_pct_adult	missing_pct_total	difference_pct
smoking_ever	85.40	90.03	-4.62
glucose	59.17	69.23	-10.06
physical_activity_freq	54.78	69.10	-14.33
doctor_visit_12m	53.89	43.71	10.18
a1c	26.38	43.73	-17.34
dbp	24.90	37.01	-12.11
sbp	24.90	37.01	-12.11
bmi	23.53	29.01	-5.49
weight_kg	23.37	26.64	-3.27
height_cm	23.19	28.78	-5.58
income_poverty_ratio	17.00	17.10	-0.10
education	4.54	34.78	-30.24
sleep_hours	1.39	29.71	-28.32
has_health_insurance	0.25	0.19	0.05
smoking_current	0.22	31.83	-31.61
diabetes_self_report	0.04	1.62	-1.58
id	0.00	0.00	0.00
age	0.00	0.00	0.00
race	0.00	0.00	0.00
gender	0.00	0.00	0.00
insurance_type	0.00	0.00	0.00

Missing Value Check

```
Missing Percentage by Variable
id                      0.00
age                     0.00
insurance_type          0.00
doctor_visit_12m        0.00
has_health_insurance    0.00
diabetes_self_report    0.00
sleep_hours             0.00
physical_activity_freq  0.00
smoking_ever            0.00
smoking_current         0.00
glucose                 0.00
a1c                     0.00
dbp                     0.00
sbp                     0.00
bmi                     0.00
height_cm               0.00
weight_kg               0.00
income_poverty_ratio    0.00
education               0.00
race                    0.00
gender                  0.00
age_bin                 0.00
dtype: float64
```

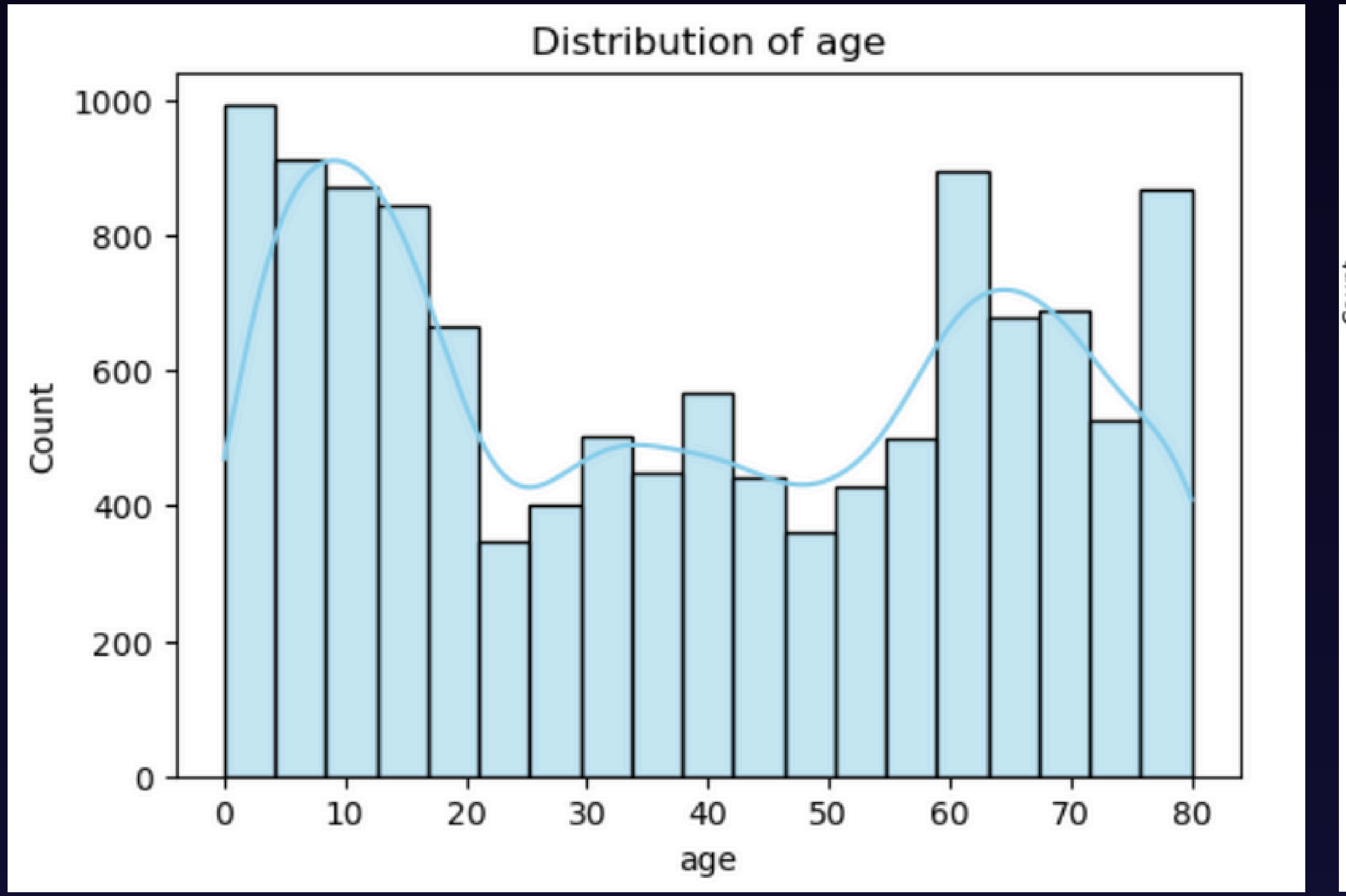
Handle the Missing Values and Outliers

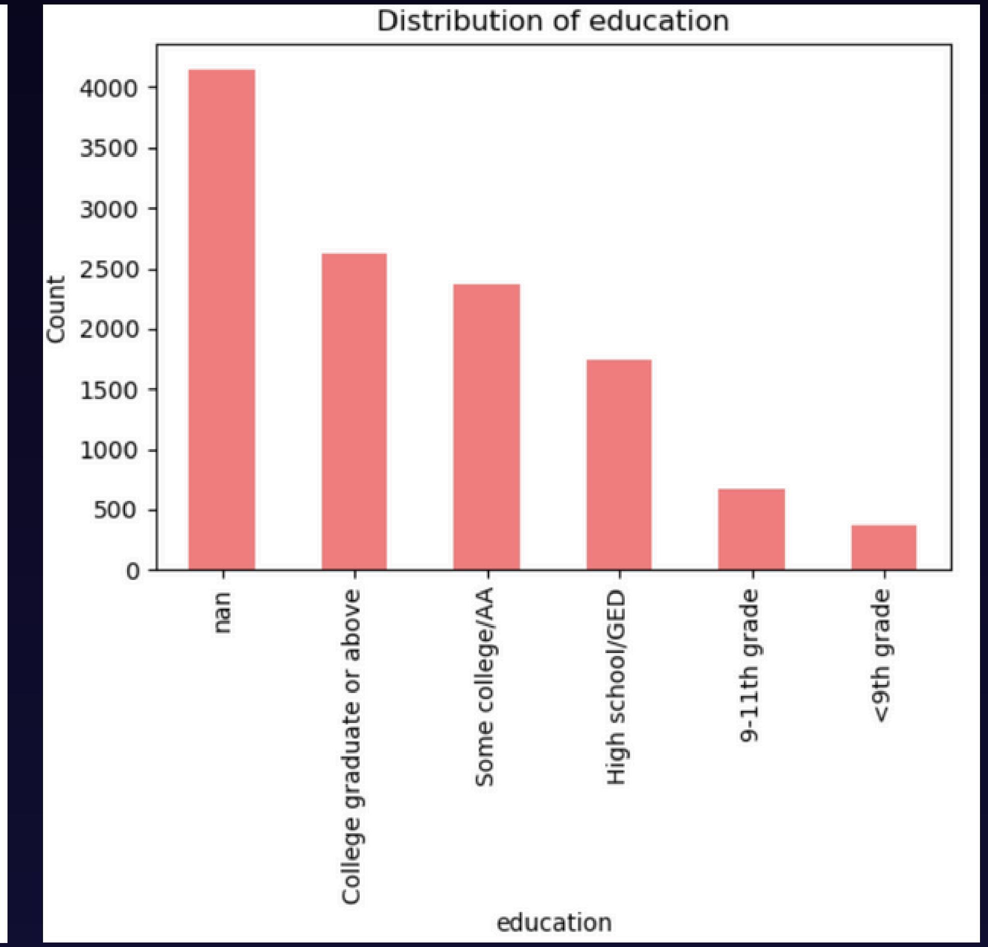
Categorical Na → "Unknown"
Numeric Na / Outlier → median imputation by_age × gender group → age group → global median

Exploratory Data Analysis (EDA)

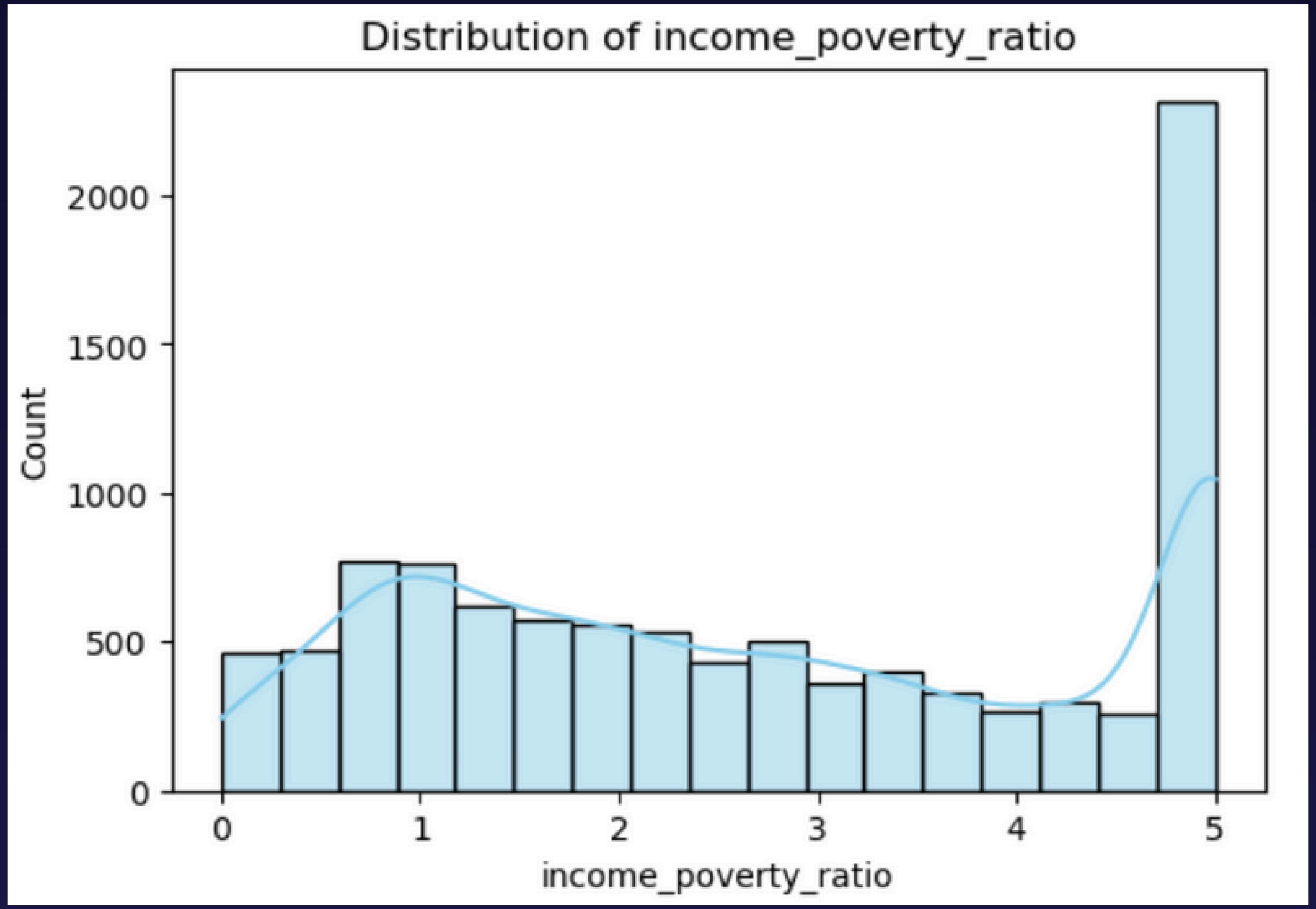
- Demographics
- Health Indicators (BMI, SBP, DBP)
- Lifestyle Factors
- Insurance Status + BP Differences Across Insurance Types

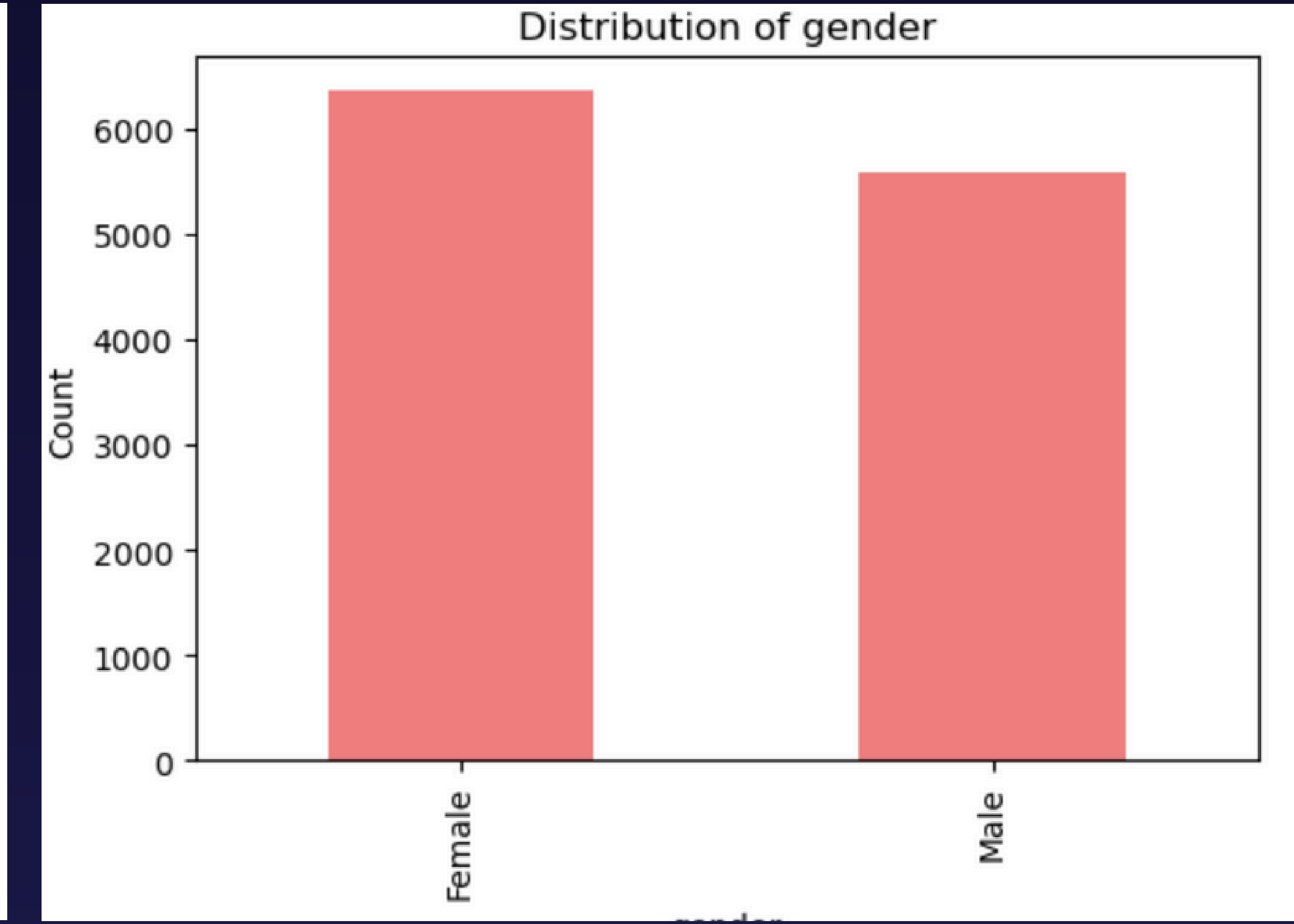
Exploratory Data Analysis (EDA)



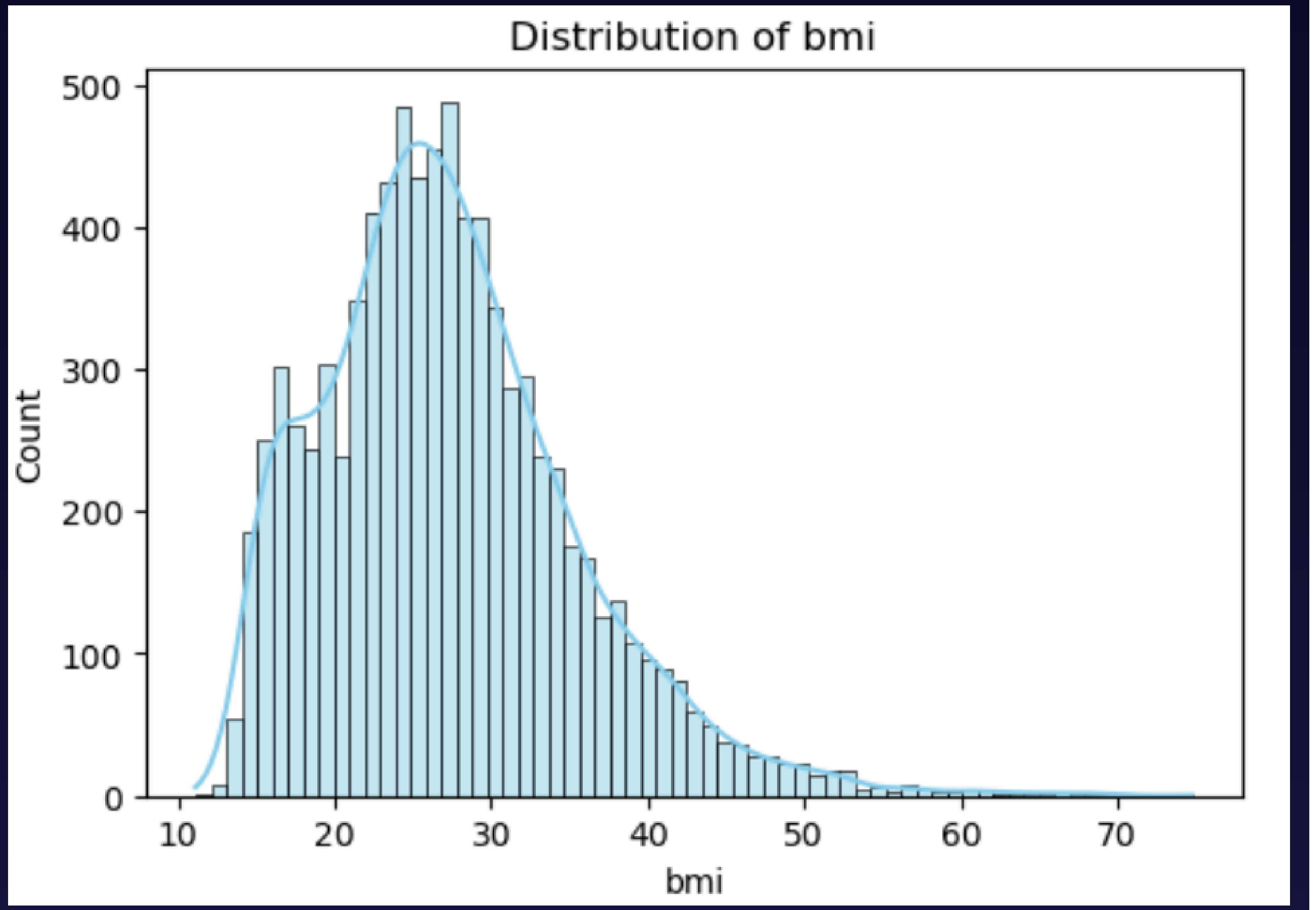


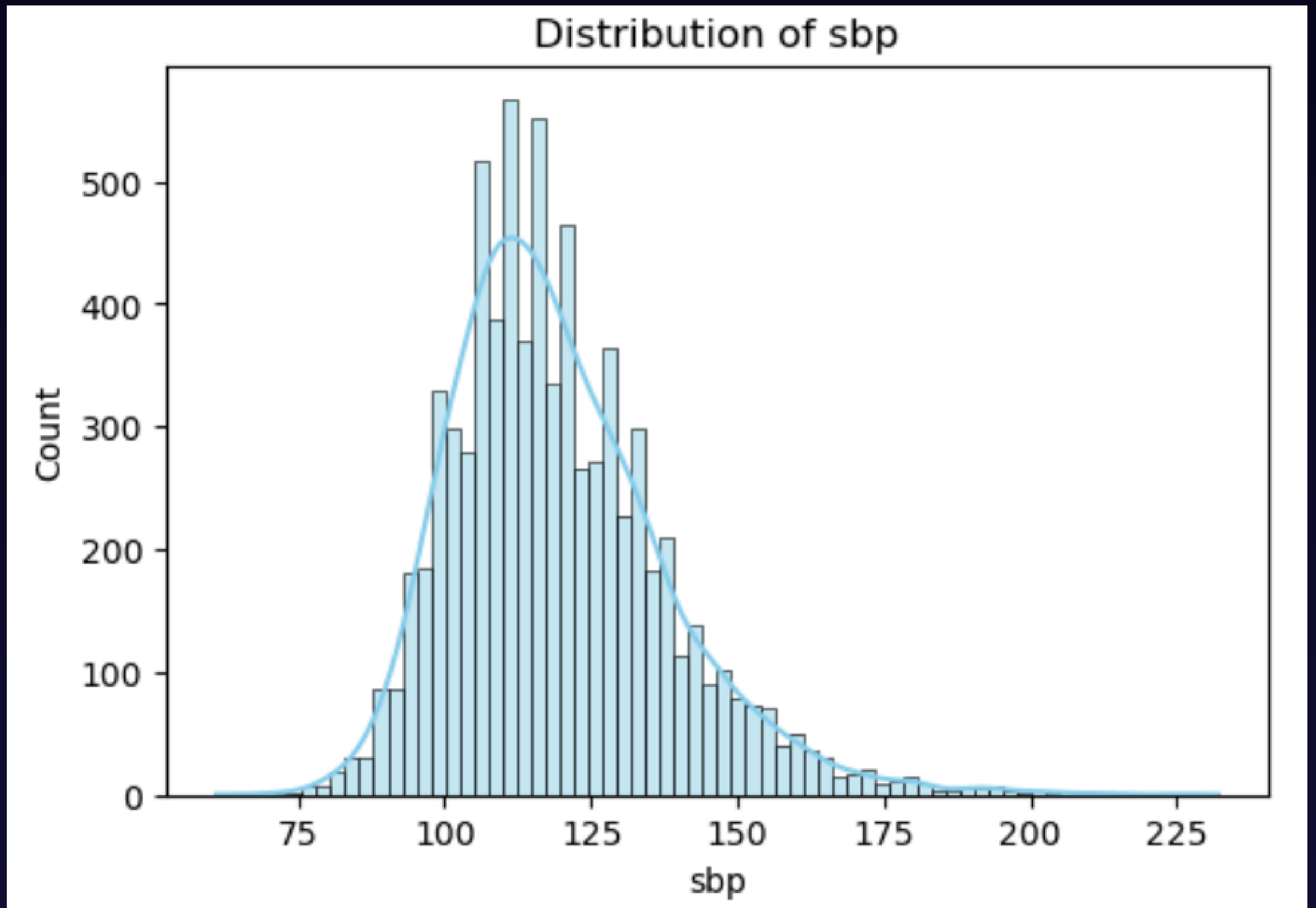
Demographics

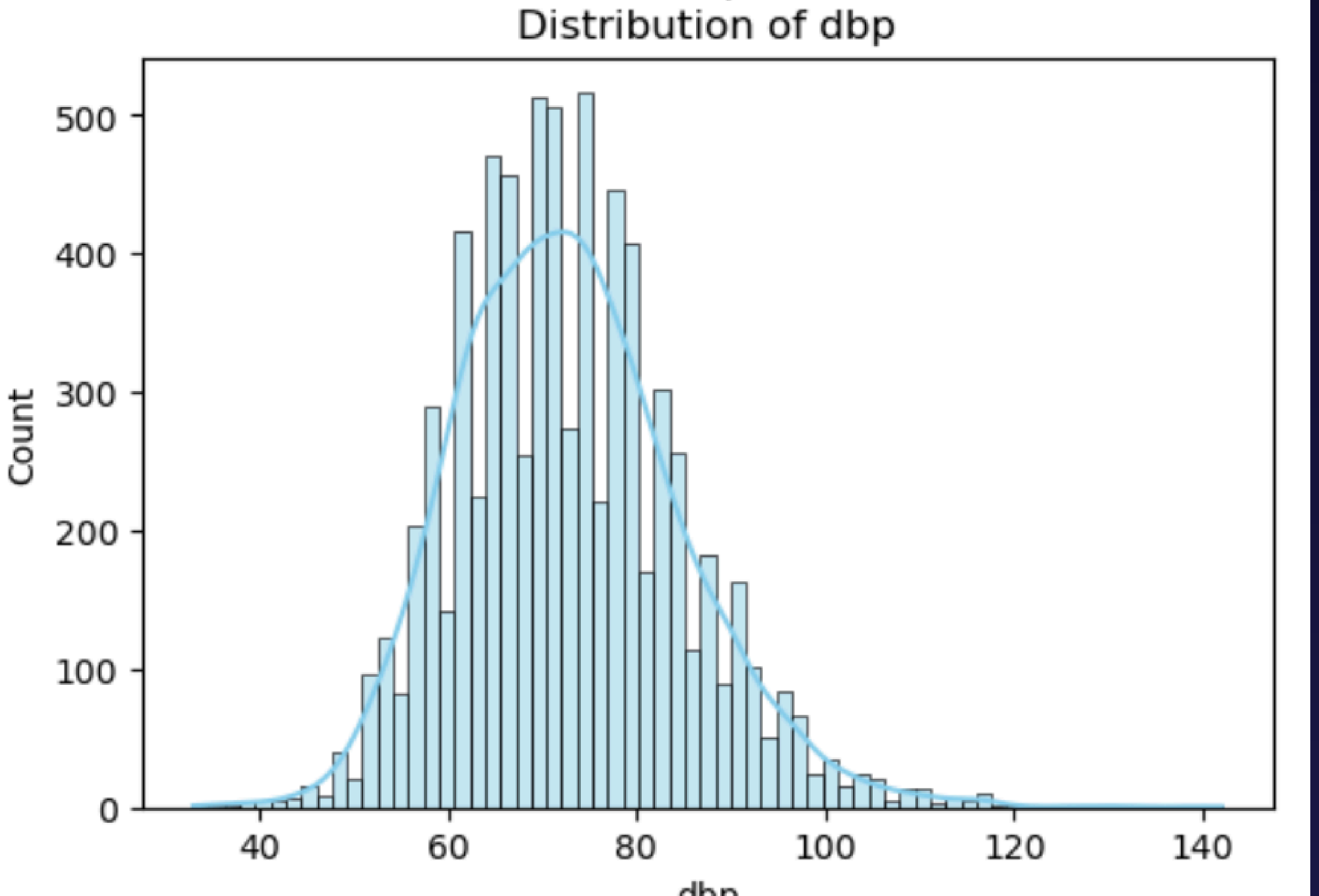




Exploratory Data Analysis (EDA)

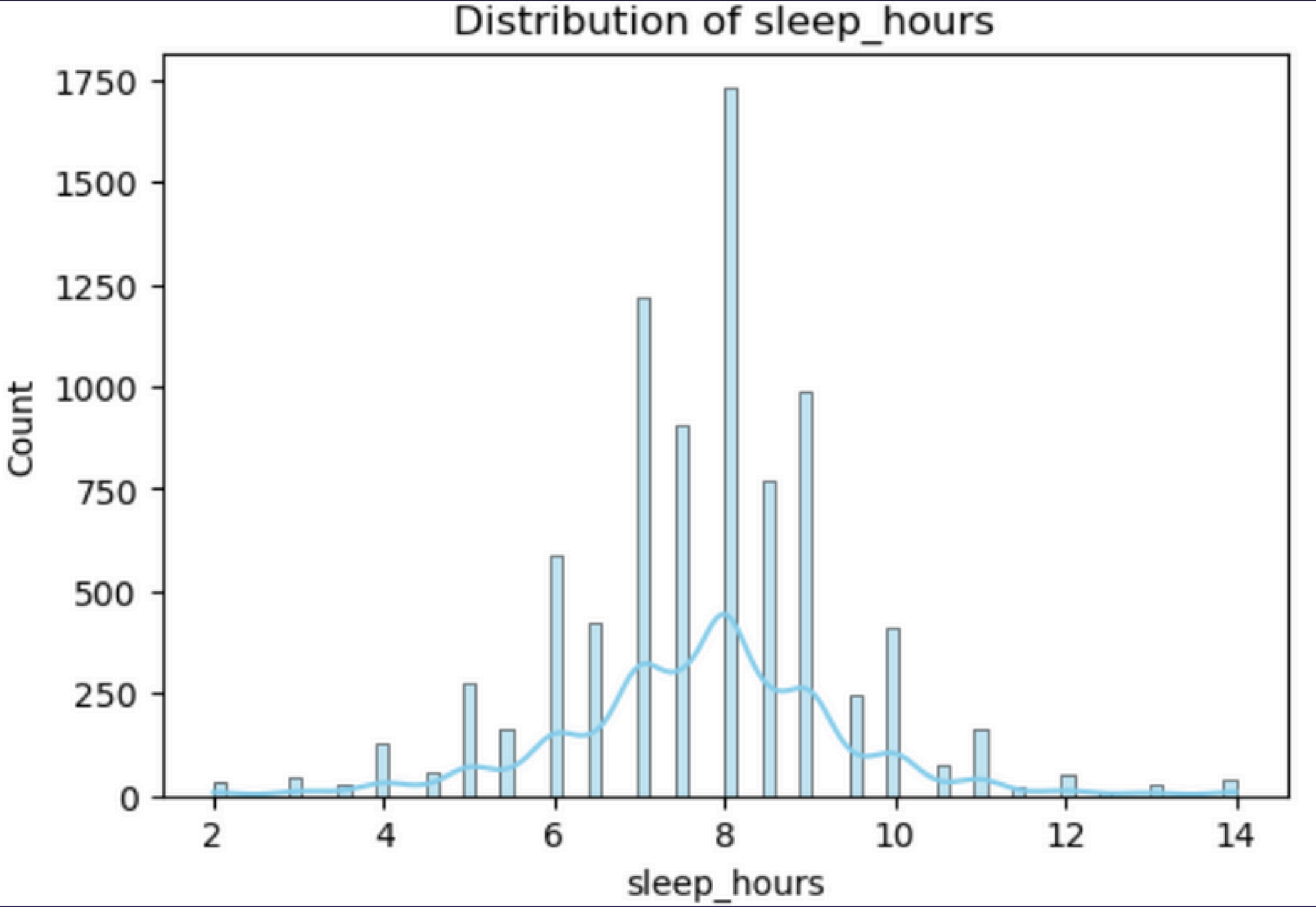


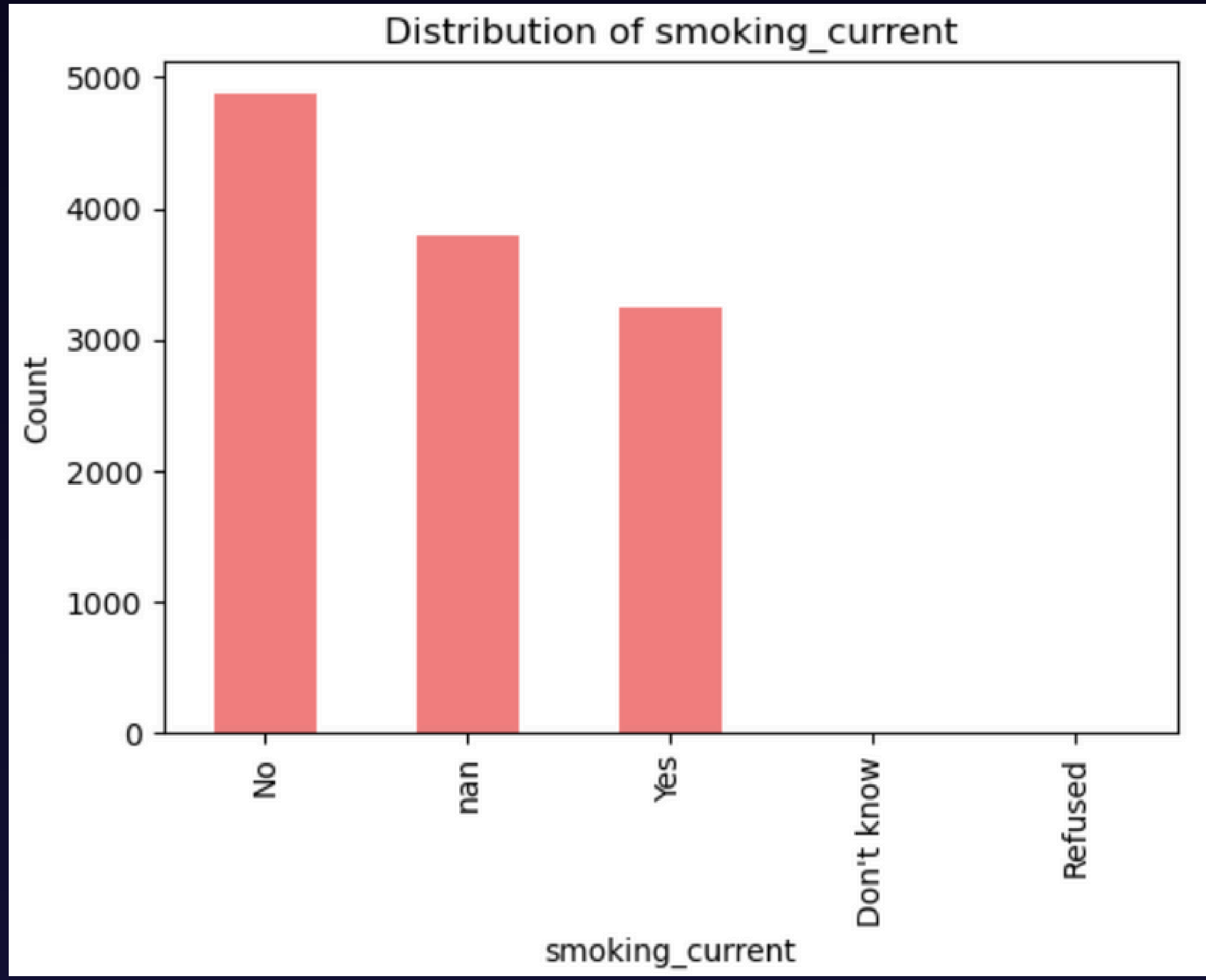


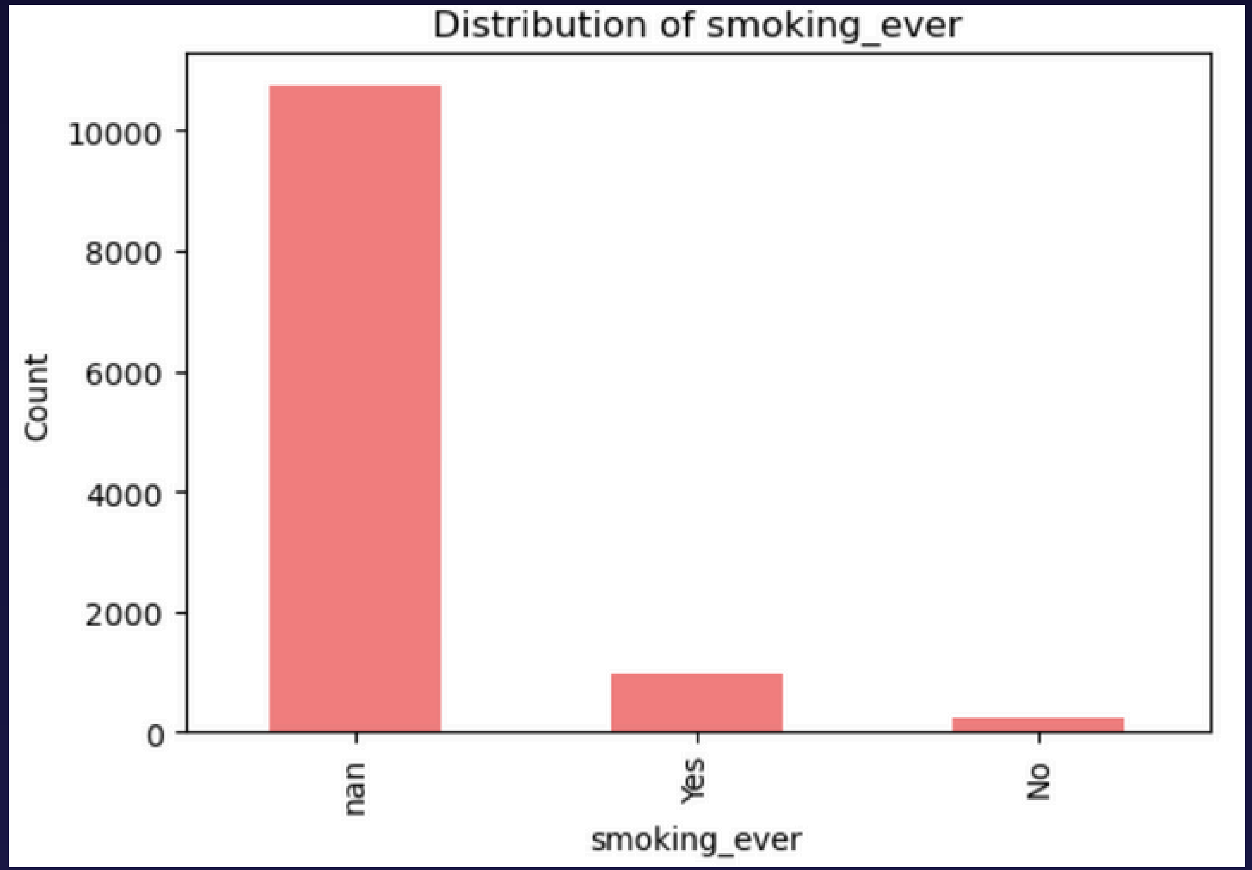


Health Indicators

Exploratory Data Analysis (EDA)

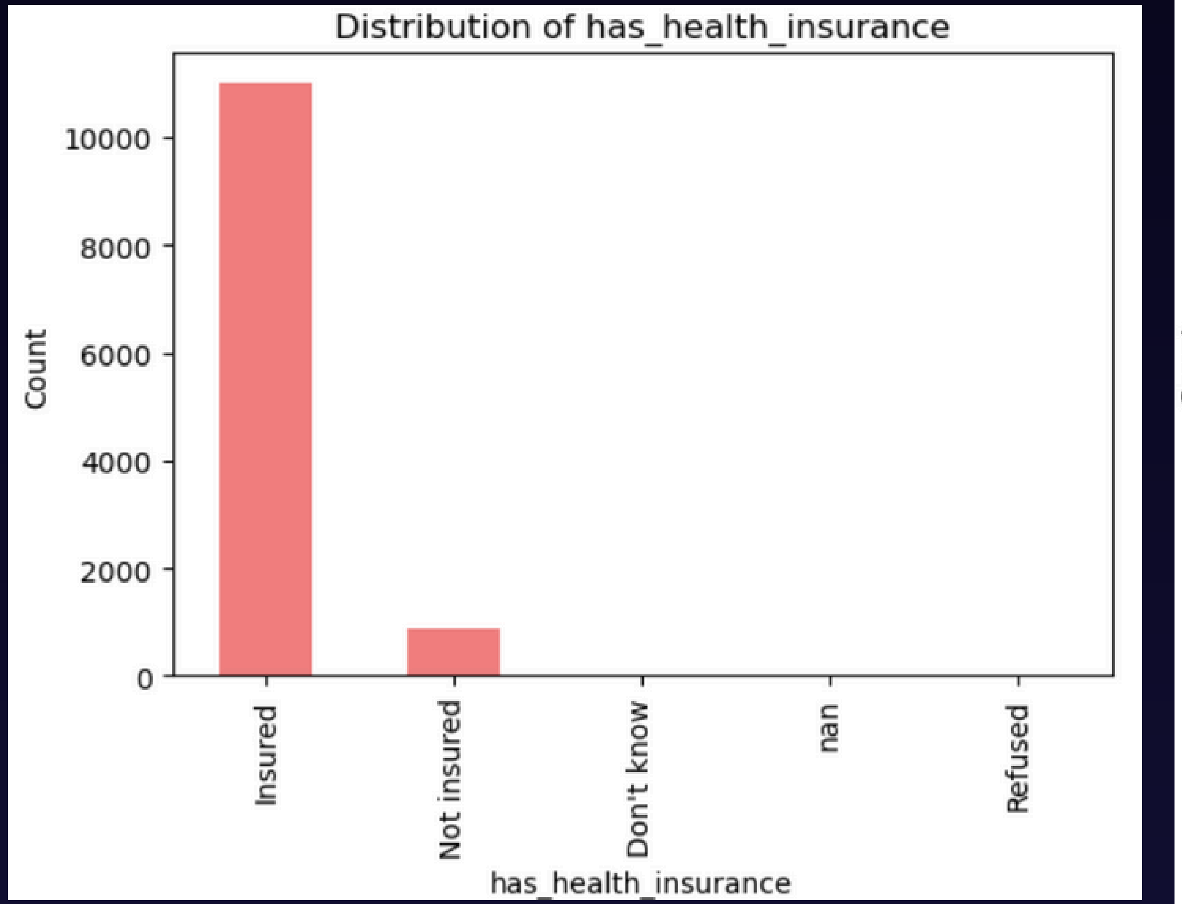


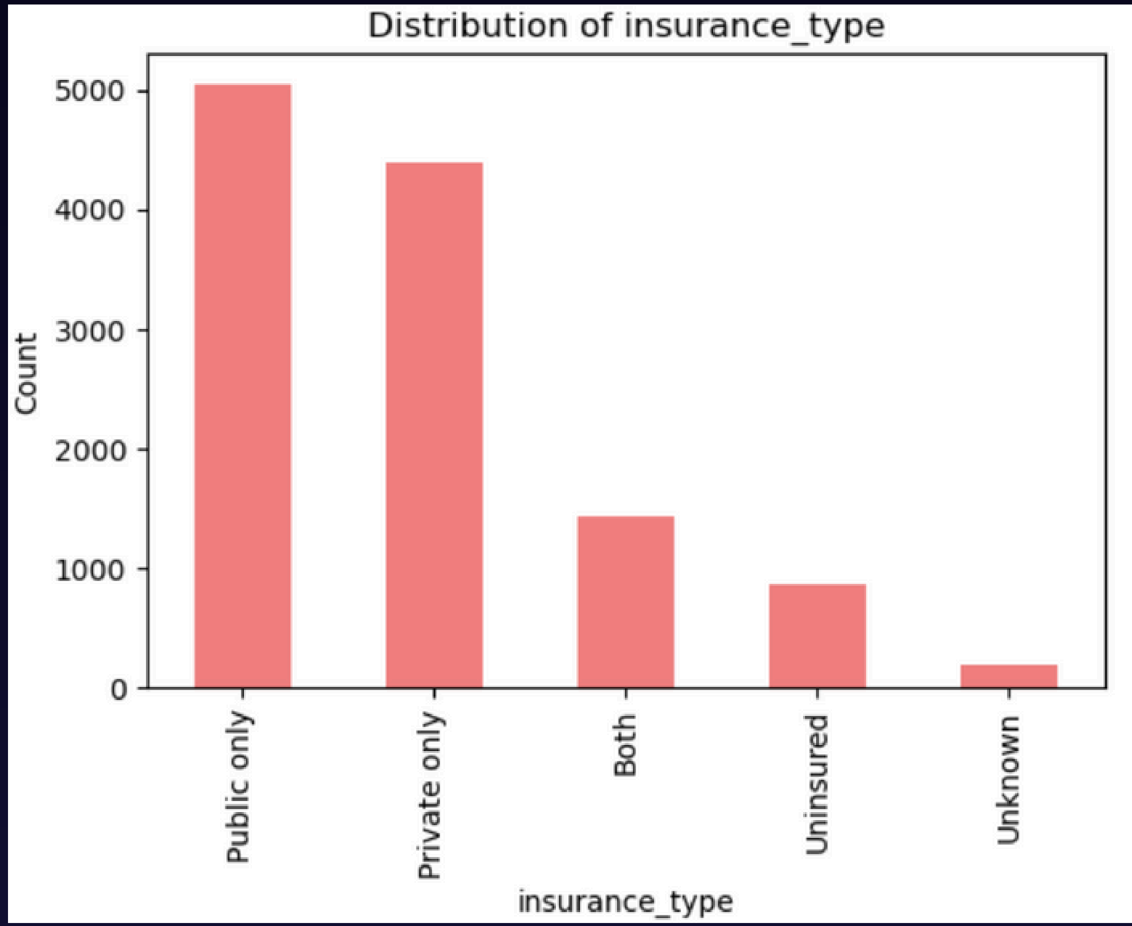


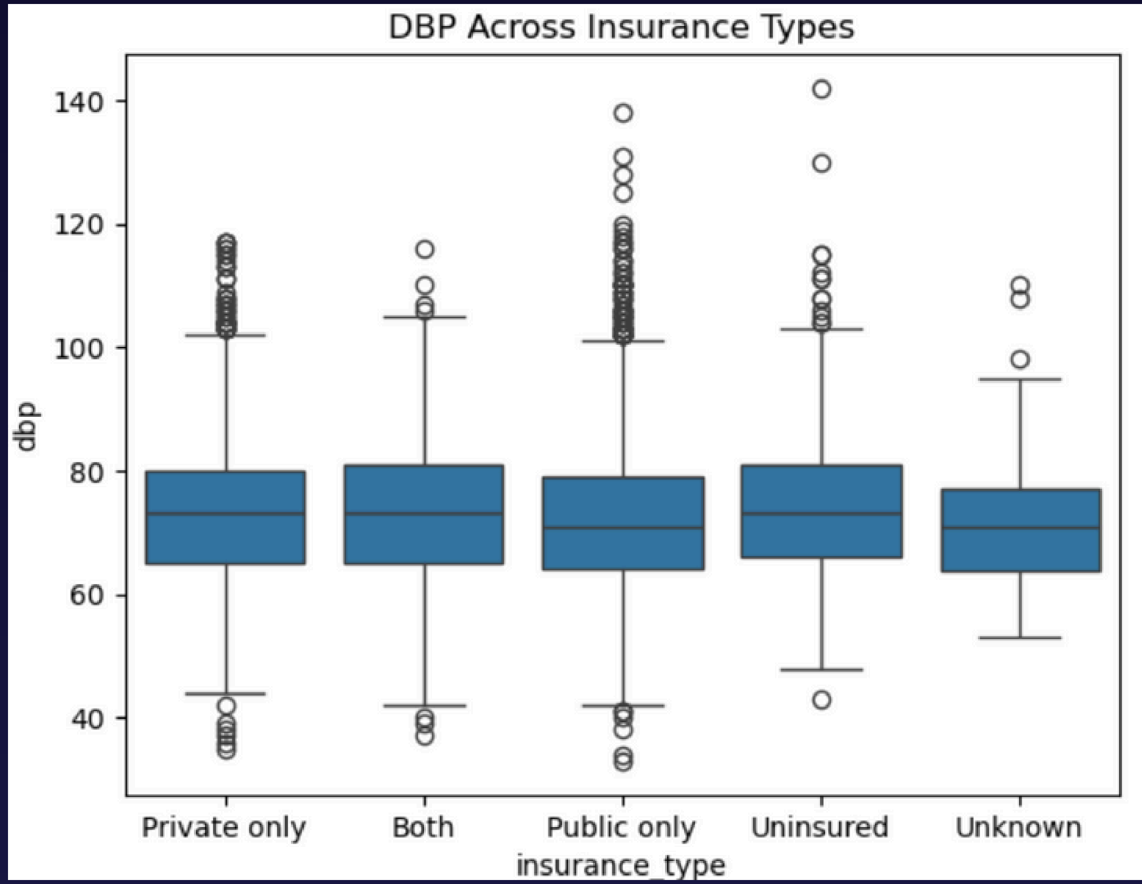


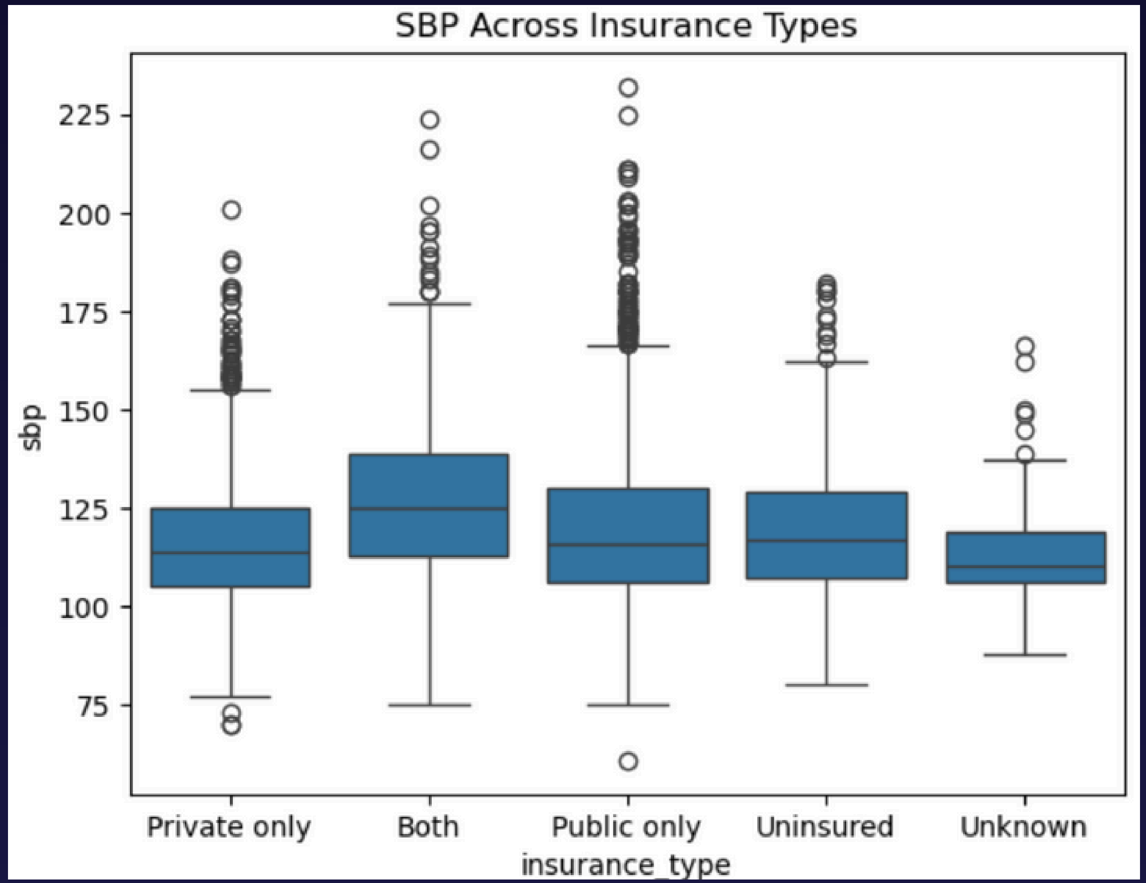
Lifestyle

Exploratory Data Analysis (EDA)









Insurance Status

Blood Pressure Differences Across Insurance Types

Association Rule Discovery: Apriori

Socio-economic → Insurance Type

	antecedents	consequents	support	confidence	lift	kulc	ir
586	(gender=Female, age_bin=61-80, race=Non-Hispanic White)	(insurance_type=Both)	0.063903	0.423922	2.695971	0.415159	0.026633
316	(age_bin=61-80, race=Non-Hispanic White)	(insurance_type=Both)	0.110879	0.406292	2.583853	0.555720	0.362274
194	(age_bin=61-80, education=College graduate or above)	(insurance_type=Both)	0.052005	0.401135	2.551058	0.365934	0.117493
136	(income_poverty_ratio_bin=>4, age_bin=31-45)	(insurance_type=Private only)	0.052864	0.874239	2.440984	0.510921	0.813883
165	(income_poverty_ratio_bin=>4, age_bin=46-60)	(insurance_type=Private only)	0.059119	0.866906	2.420510	0.515987	0.789579
112	(age_bin=31-45, education=College graduate or above)	(insurance_type=Private only)	0.068318	0.816716	2.280370	0.503734	0.734975
149	(education=College graduate or above, age_bin=46-60)	(insurance_type=Private only)	0.053968	0.791367	2.209594	0.471026	0.778656
143	(age_bin=31-45, race=Non-Hispanic White)	(insurance_type=Private only)	0.069668	0.664327	1.854884	0.429424	0.643904
650	(income_poverty_ratio_bin=>4, gender=Female, education=College graduate or above)	(insurance_type=Private only)	0.059978	0.655496	1.830226	0.411481	0.684293
178	(race=Non-Hispanic White, age_bin=46-60)	(insurance_type=Private only)	0.077885	0.647299	1.807338	0.432382	0.593693
366	(income_poverty_ratio_bin=>4, education=College graduate or above)	(insurance_type=Private only)	0.109776	0.633853	1.769795	0.470180	0.438755
463	(gender=Female, income_poverty_ratio_bin=<=1 (Below poverty line))	(insurance_type=Public only)	0.054949	0.663704	1.733240	0.403601	0.730666
715	(income_poverty_ratio_bin=>4, education=College graduate or above, race=Non-Hispanic White)	(insurance_type=Private only)	0.080216	0.614085	1.714600	0.419029	0.556890
468	(income_poverty_ratio_bin=>4, gender=Female)	(insurance_type=Private only)	0.086594	0.604452	1.687705	0.423116	0.518037
99	(income_poverty_ratio_bin=<=1 (Below poverty line))	(insurance_type=Public only)	0.089415	0.641725	1.675845	0.437615	0.562766

1. Rules leading to “Both” (dual coverage):
Older adults such as 'age 61–80' who combine *Medicare* eligibility with private supplemental plans

2. Rules leading to “Private only” :
Middle-aged adults with **high-income** or **college-education**

3. Rules leading to “Public only”:
Low-income population

Association Rule Discovery: Apriori

Multi-Domain Determinant → Hypertension

	antecedents	consequents	support	confidence	lift	kulc	ir
1402	(bmi_bin=Obesity, race=Non-Hispanic White)	(htn_stage=Stage 1)	0.063535	0.329098	1.463794	0.305847	0.089650
1412	(bmi_bin=Obesity, smoking_current=No)	(htn_stage=Stage 1)	0.062799	0.324051	1.441345	0.301687	0.087211
100	(bmi_bin=Obesity)	(htn_stage=Stage 1)	0.107200	0.323464	1.438736	0.400139	0.237367
1408	(sleep_bin=Normal (6-8h), bmi_bin=Obesity)	(htn_stage=Stage 1)	0.053845	0.318347	1.415974	0.278922	0.163722
2866	(age_bin=18-30, bmi_bin=Overweight, smoking_current=No)	(htn_stage=Normal)	0.052251	0.940397	1.386950	0.508730	0.913591
322	(age_bin=18-30, bmi_bin=Overweight)	(htn_stage=Normal)	0.067705	0.940375	1.386917	0.520115	0.888190
310	(age_bin=18-30, bmi_bin=Normal)	(htn_stage=Normal)	0.054336	0.911523	1.344364	0.495830	0.905044
332	(age_bin=18-30, gender=Female)	(htn_stage=Normal)	0.082056	0.910204	1.342419	0.515612	0.856811
2878	(age_bin=18-30, gender=Female, smoking_current=No)	(htn_stage=Normal)	0.071630	0.908243	1.339526	0.506943	0.874351
2930	(age_bin=31-45, gender=Female, bmi_bin=Overweight)	(htn_stage=Normal)	0.050902	0.908096	1.339311	0.491584	0.910413

	antecedents	consequents	support	confidence	lift	kulc	ir
58	(age_bin=61-80)	(htn_stage=Stage 2)	0.059242	0.145701	1.499878	0.377775	0.696192
1402	(bmi_bin=Obesity, race=Non-Hispanic White)	(htn_stage=Stage 1)	0.063535	0.329098	1.463794	0.305847	0.089650
1412	(bmi_bin=Obesity, smoking_current=No)	(htn_stage=Stage 1)	0.062799	0.324051	1.441345	0.301687	0.087211
100	(bmi_bin=Obesity)	(htn_stage=Stage 1)	0.107200	0.323464	1.438736	0.400139	0.237367
1408	(sleep_bin=Normal (6-8h), bmi_bin=Obesity)	(htn_stage=Stage 1)	0.053845	0.318347	1.415974	0.278922	0.163722
1336	(bmi_bin=Obesity, gender=Female)	(htn_stage=Stage 1)	0.059119	0.291239	1.295400	0.277098	0.059215
38	(age_bin=46-60)	(htn_stage=Stage 1)	0.057770	0.271470	1.207470	0.264213	0.031644
984	(age_bin=61-80, smoking_current=No)	(htn_stage=Stage 1)	0.056789	0.268873	1.195922	0.260732	0.035899
974	(age_bin=61-80, race=Non-Hispanic White)	(htn_stage=Stage 1)	0.073347	0.268764	1.195435	0.297503	0.113295

1. Risk-increasing patterns

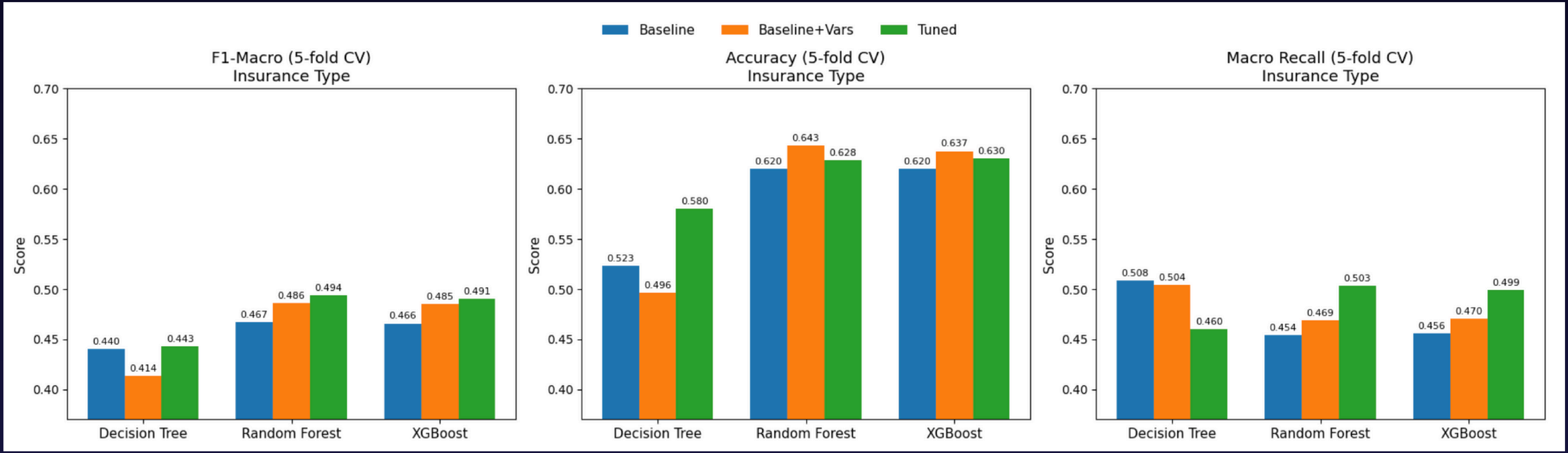
Obesity is one of **the earliest detectable risk markers**. This suggests that early **weight-management interventions** could substantially delay or prevent the transition from Normal BP to Stage 1.

2. Age-driven patterns

Age is one of the strongest protective buffer . Young age protects, obesity pushes individuals into Stage 1, and **aging** ultimately drives progression to **Stage 2**.

Classification for Insurance Type Model Comparison

*Decision Tree, 👍**Random Forest**, XGBoost*



Optimzation Steps:

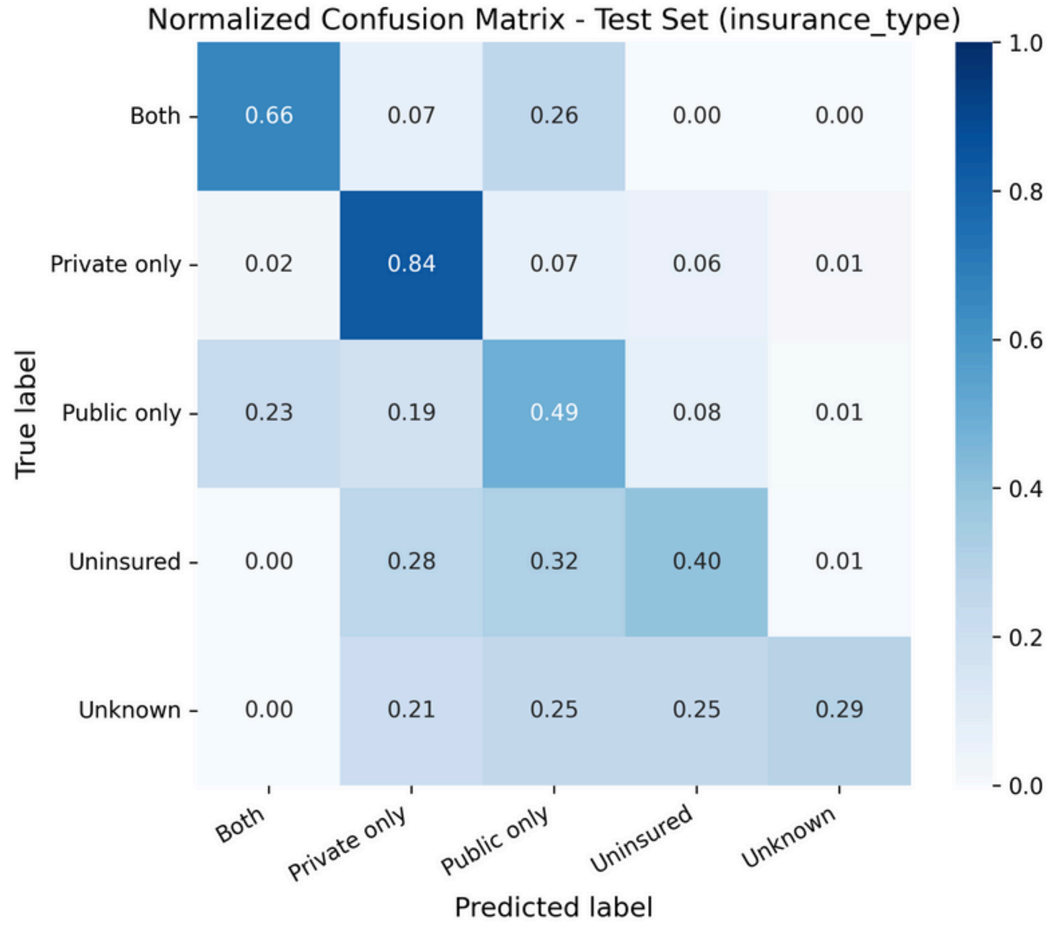
- **Feature Expansion:** marital_status, household_size, is_employed
- **Stratified Cross Valiation**
- **Oversampling** to address class imbalance
- **Hyperparameter Tuning: Macro-F1** optimization

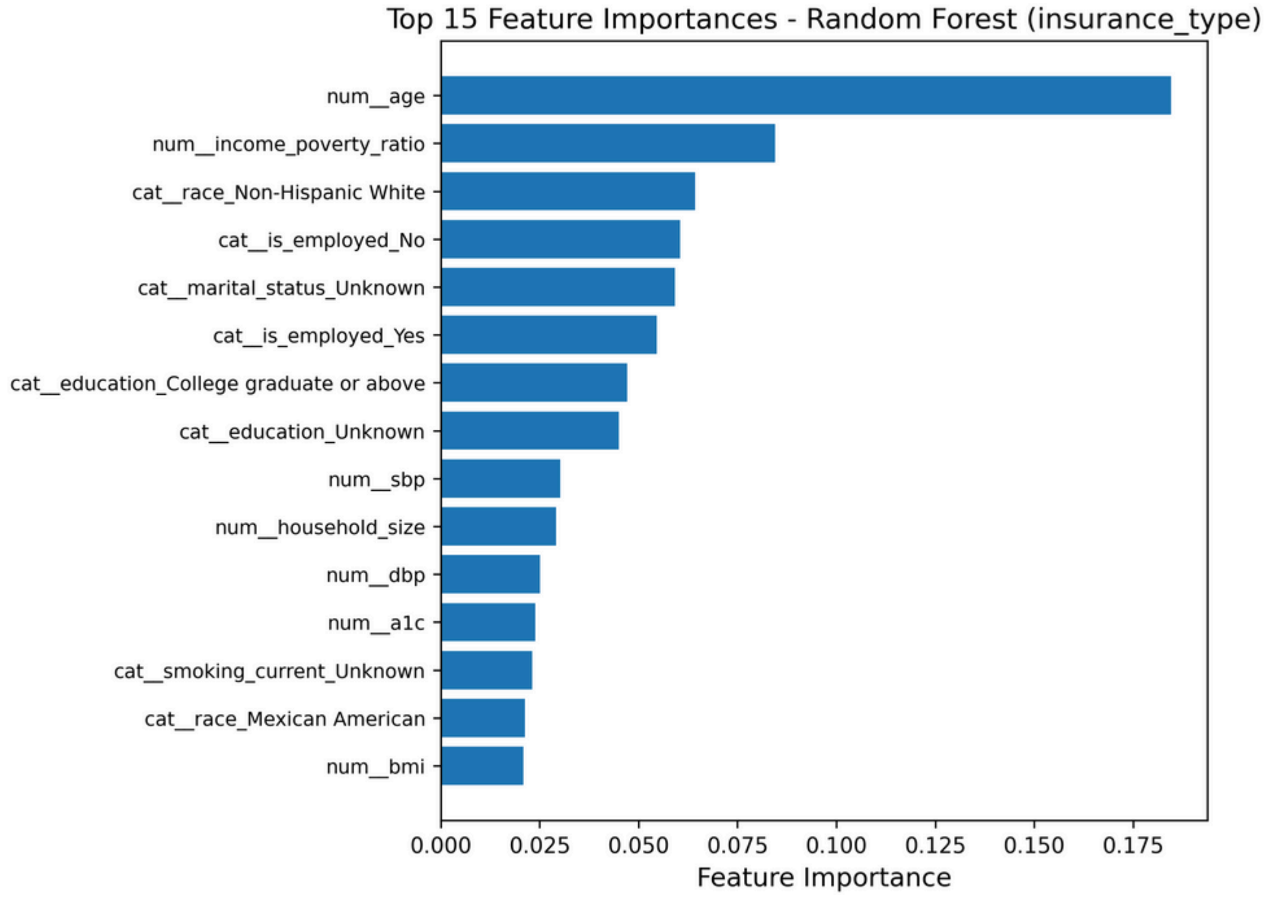
Classification for Insurance Type Model Evaluation

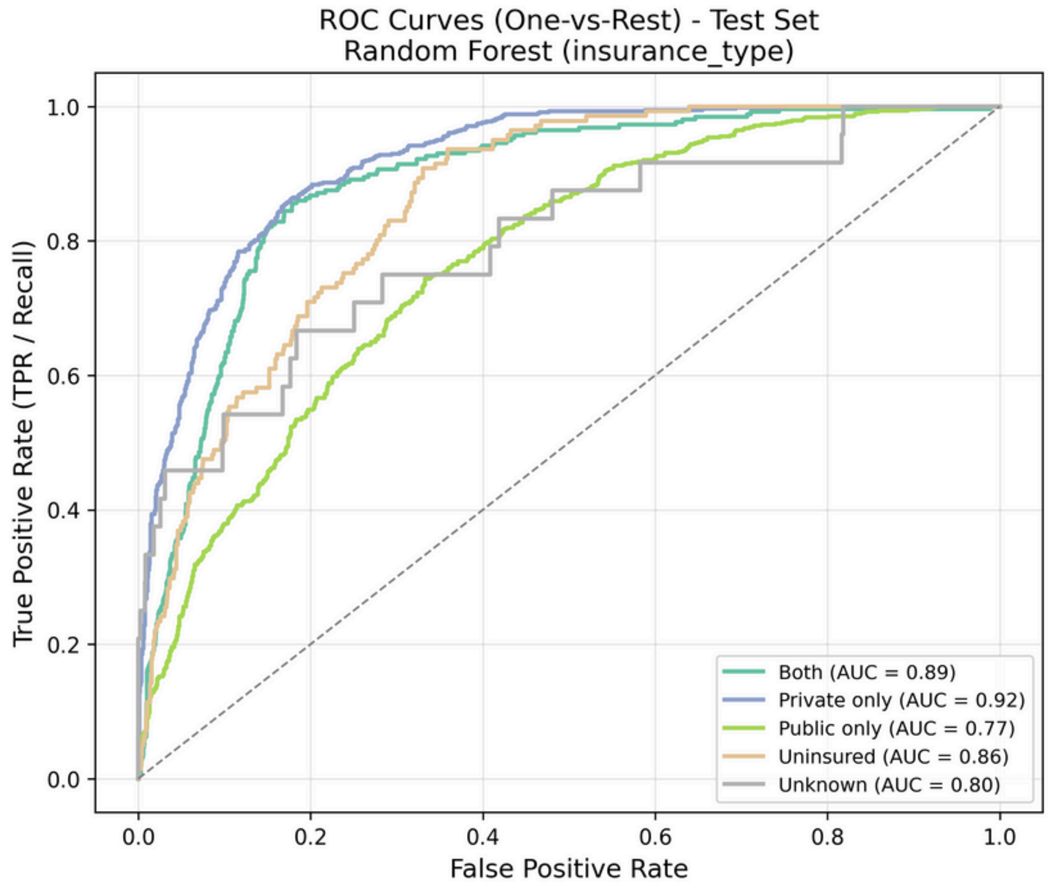
→

FINAL MODEL PERFORMANCE ON TEST SET (insurance_type)

	precision	recall	f1-score	support
Both	0.52	0.66	0.58	257
Private only	0.73	0.84	0.78	584
Public only	0.66	0.49	0.56	625
Uninsured	0.38	0.40	0.39	141
Unknown	0.33	0.29	0.31	24
accuracy			0.63	1631
macro avg	0.52	0.54	0.53	1631
weighted avg	0.63	0.63	0.63	1631







- **Identifies the majority of Private-only (84%) and Both (66%) cases**
- Most **confusion occurs** between **Public-only** vs. Both and **Uninsured** vs. Public-only

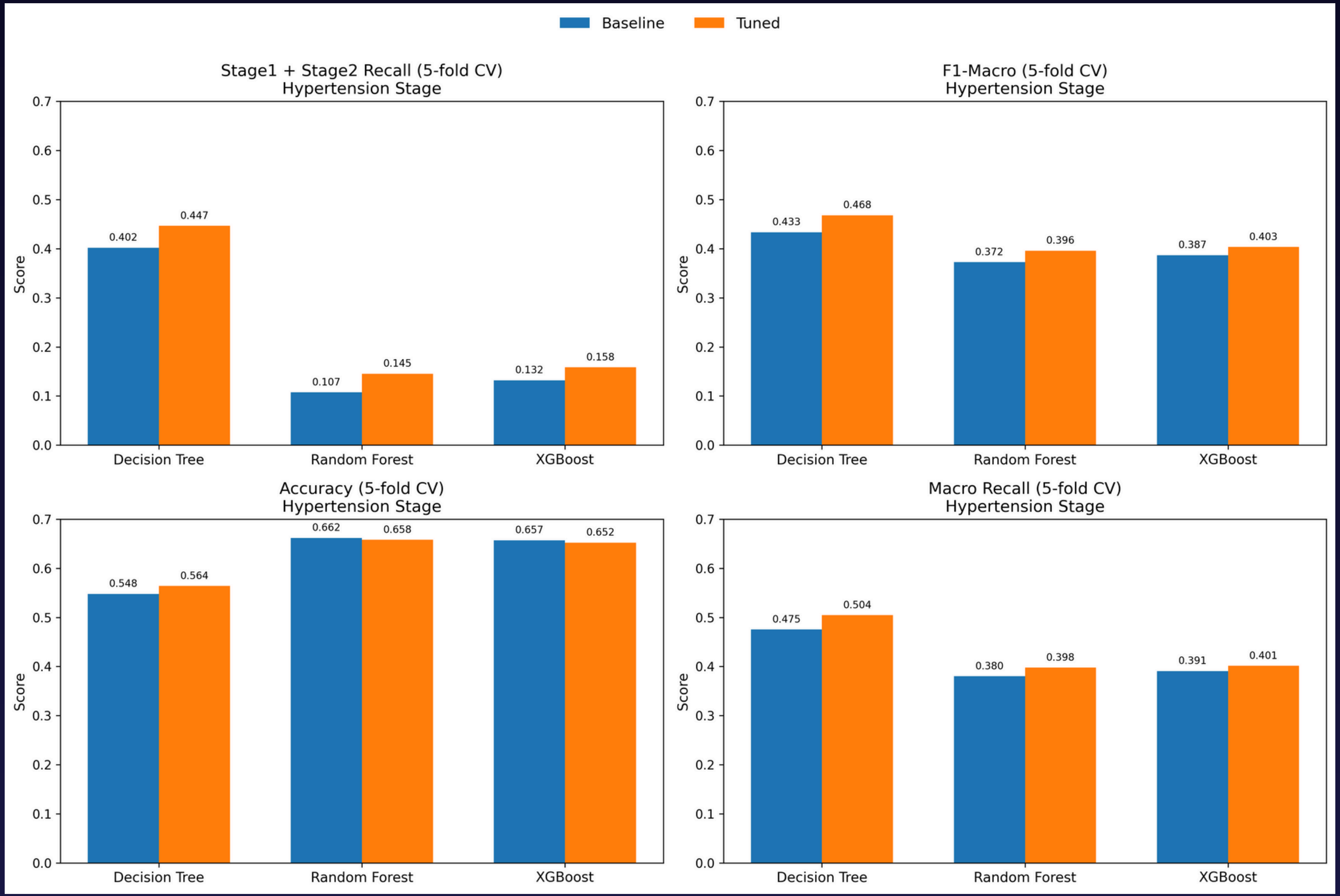
- **Age and income–poverty ratio dominate prediction**
- **Race, employment, and education** collectively form **the next tier**
- **Clinical measures** (BP, A1C) **play a smaller role**

- Overall model performance is strong
- AUC scores are **high for Private-only (0.92) and Both (0.89)**

Classification for Hypertension Risk Prediction

Model Comparison

👍***Decision Tree***, *Random Forest, XGBoost*



Define hypertension Classification Labels:

```python
def hypertension_stage(row):
    sbp = row['sbp']
    dbp = row['dbp']

    # Normal (incl. Elevated)
    if sbp < 130 and dbp < 80:
        return "Normal"

    # Stage 1 Hypertension
    elif (130 <= sbp < 140) or (80 <= dbp < 90):
        return "Stage 1"

    # Stage 2 Hypertension
    elif sbp >= 140 or dbp >= 90:
        return "Stage 2"

    else:
        return "Undefined"
```

Optimzation Steps:

- **Stratified Cross Valiation**
- **Oversampling** to address class imbalance
- **Hyperparameter Tuning: Aveage Recall of Stage 1 + Stage 2 as** the tailored tuning indicator.

Best Model: Decison Tree:

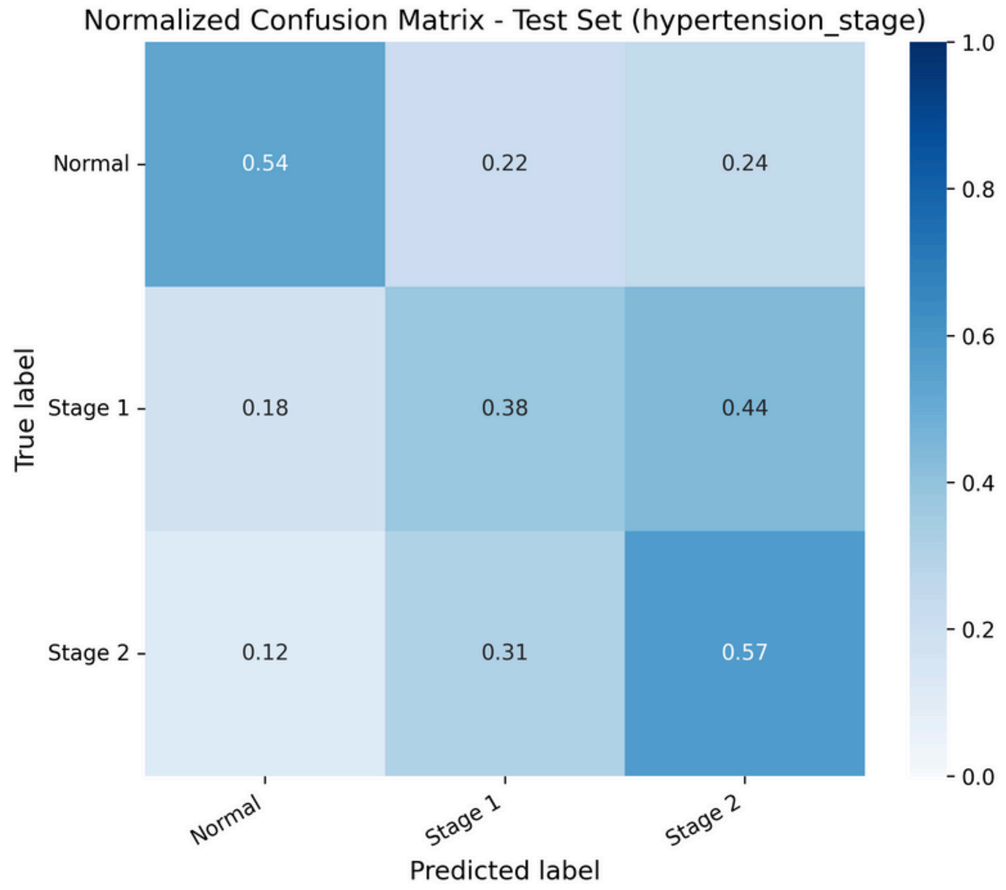
- **Clear threshold patterns: Well-defined cutoffs** in variables, such as age, BMI.
- **Strong rule-based separability - Hypertension stage criteria**

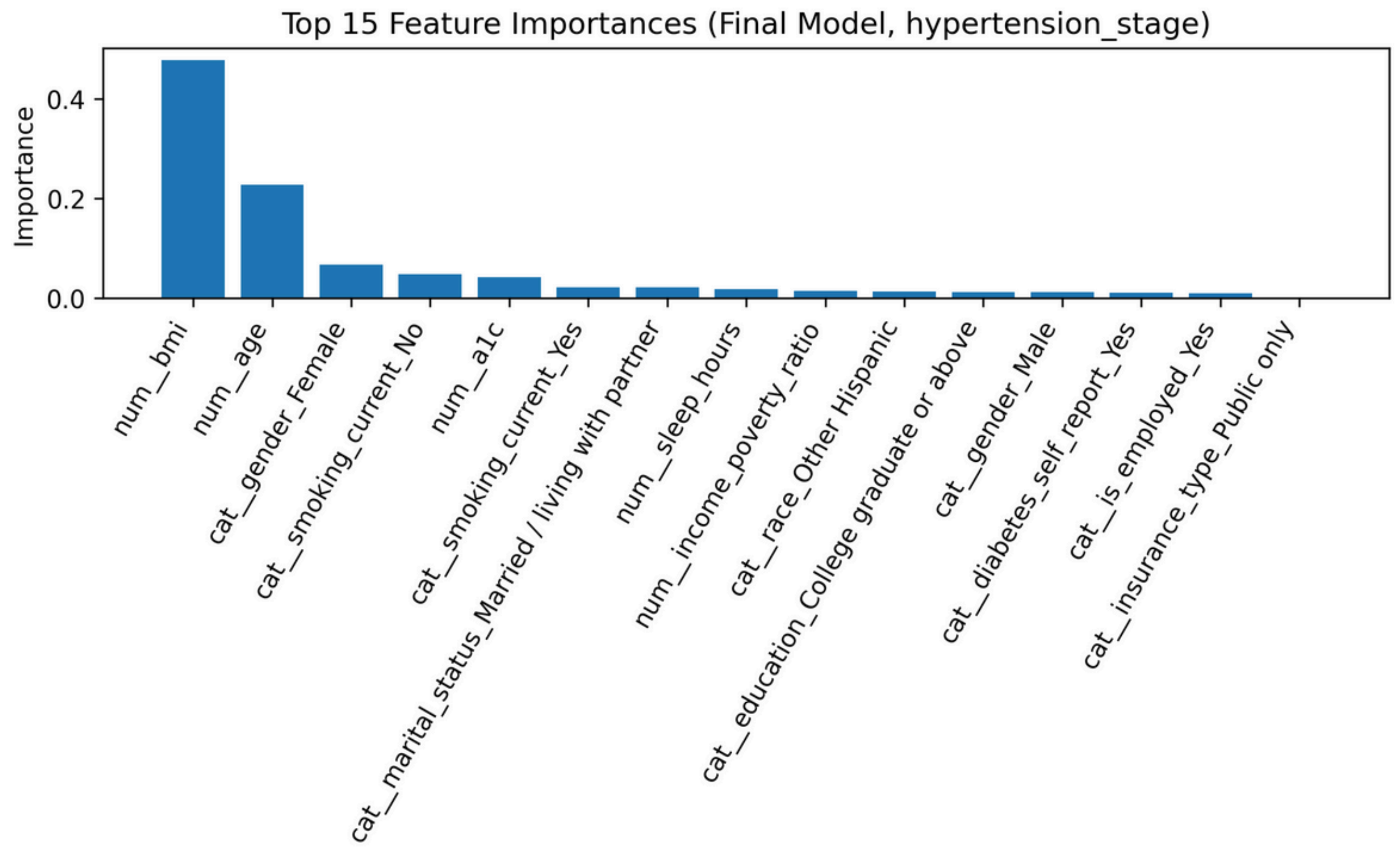
Classification for Hypertension Risk Prediction
Model Evaluation

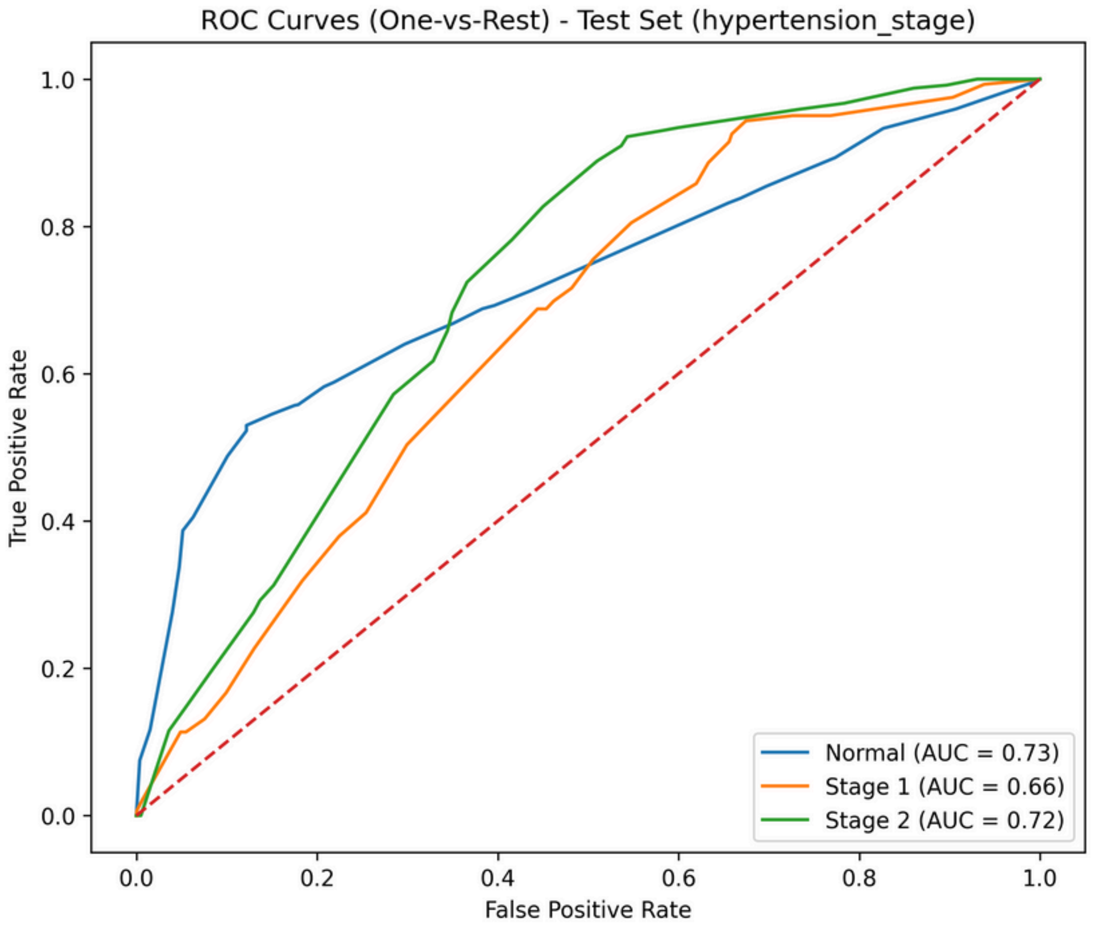
→

FINAL MODEL PERFORMANCE ON TEST SET (hypertension_stage)

	precision	recall	f1-score	support
Normal	0.88	0.54	0.67	1106
Stage 1	0.25	0.38	0.30	282
Stage 2	0.26	0.57	0.36	243
accuracy			0.52	1631
macro avg	0.47	0.50	0.44	1631
weighted avg	0.68	0.52	0.56	1631



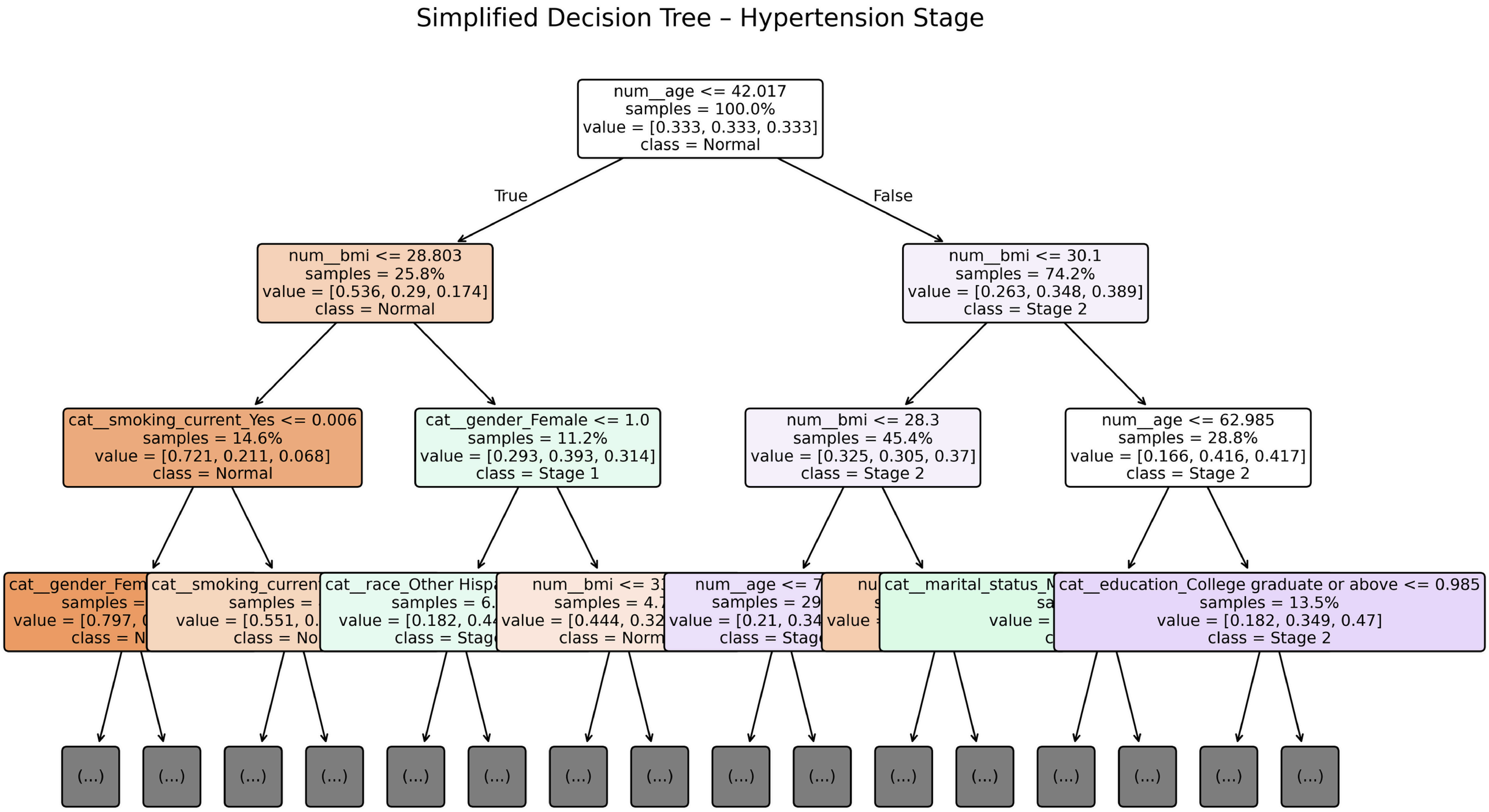




- Identifies **54% of Normal cases**
- **Stage 2 is recovered relatively well (57% recall)**, showing the model captures the more **severe hypertension patterns** aligned with **clearer signals** (older Age, higher BMI)
- **Stage 1 exhibits the most confusion**

- **BMI and Age overwhelmingly dominate prediction**
- **Behavioral factors are additional predictive signal**
- **Socioeconomic variables** play **a secondary role** vs direct physiological indicators.
- **Insurance type** contributes **minimally**

- **Normal** and **Stage 2** are more **distinguishable**.
- Stage 1 has a lower AUC reflecting real-world difficulty: **mild hypertension is the hardest to clinically identify**


Simplified Decision Tree – Hypertension Stage
num__age <= 42.017
samples = 100.0%
value = [0.333, 0.333, 0.333]
class = Normal
True
False
num__bmi <= 28.803
samples = 25.8%
value = [0.536, 0.29, 0.174]
class = Normal
num__bmi <= 30.1
samples = 74.2%
value = [0.263, 0.348, 0.389]
class = Stage 2
cat__smoking_current_Yes <= 0.006
samples = 14.6%
value = [0.721, 0.211, 0.068]
class = Normal
cat__gender_Female <= 1.0
samples = 11.2%
value = [0.293, 0.393, 0.314]
class = Stage 1
num__bmi <= 28.3
samples = 45.4%
value = [0.325, 0.305, 0.37]
class = Stage 2
num__age <= 62.985
samples = 28.8%
value = [0.166, 0.416, 0.417]
class = Stage 2
cat__education_College graduate or above <= 0.985
samples = 13.5%
value = [0.182, 0.349, 0.47]
class = Stage 2
(...)

Clustering

Why? →
- Go beyond prediction → Understand population subgroups
- Identify high-risk hypertensive groups + uninsured groups

What Method? →

K-Prototypes

- Mixed data : numeric + categorical
- K-Prototypes (Euclidean + matching dissimilarity)

How Many Clusters? →

k = 4

- Elbow method
- Clinical interpretability (HTN severity)
- Balanced cluster size

Cluster Evaluation

Silhouette Score: 0.075912684

Expectedly low due to **mixed-type high-dimensional data**,but still indicates weak but valid separation among clusters.

Intra-cluster variance:

cluster	age	income_poverty_ratio	bmi	sbp	dbp
0	0.294453	0.517300	0.594065	0.638280	0.582088
1	0.426346	0.601847	0.900614	0.346528	0.537153
2	0.680614	0.293532	0.587659	0.468653	0.549163
3	0.534743	0.904916	1.900174	1.349725	1.069494

cluster	a1c	sleep_hours	household_size
0	0.817507	3.088440	0.396888
1	0.185859	2.549380	1.084883
2	0.240372	1.630379	0.445144
3	3.215593	3.094341	0.841527

- **Cluster 1: lowest variance**
  young, low-risk,behaviorally homogeneous group
- **Cluster 3: highest variance**
  older population,broad range of hypertension severity
- **Clusters 0 and 2: moderate variance**
  mixed SES and mixed insurance groups

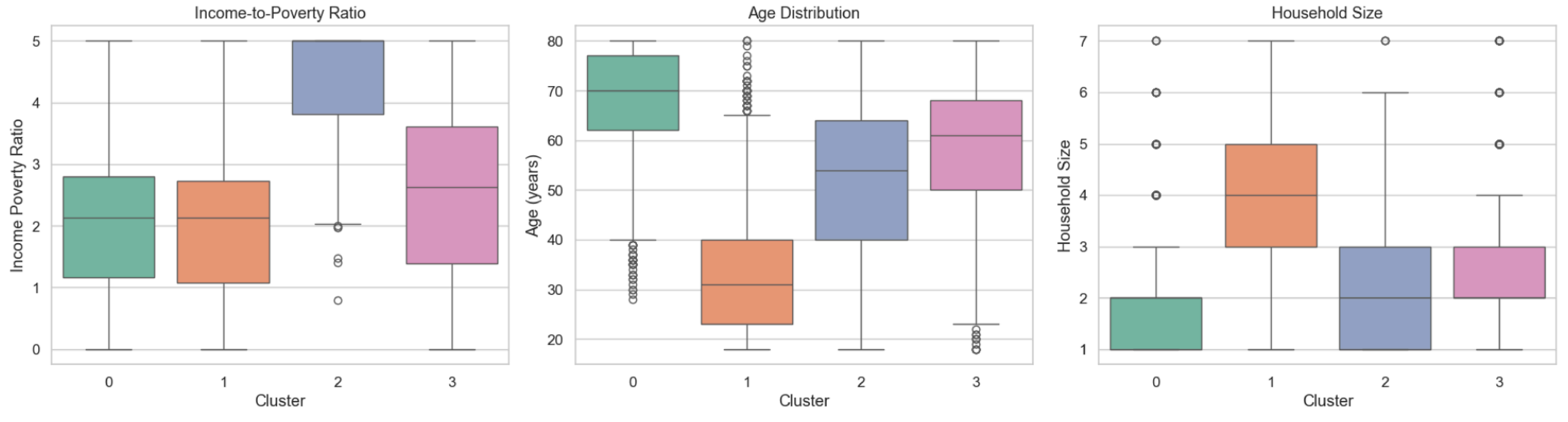
Cluster Evaluation

```
Cluster centroids:
[['0.8730199986532764' '-0.5121074944034377' '-0.08736015493669068'
  '0.1444354557854423' '-0.3121015096052984' '0.20646308518700207'
  '0.2665377527818011' '-0.5407949408788704' 'Female'
  'Non-Hispanic White' 'High school/GED' 'Widowed / divorced / separated'
  'No' 'Yes']
 ['-1.030339790388816' '-0.5380371393951842' '-0.19446126240629147'
  '-0.6544936197366779' '-0.3991135936507207' '-0.3914089901206005'
  '0.021810082373742684' '0.8934193653737671' 'Female'
  'Non-Hispanic White' 'Some college/AA' 'Married / living with partner'
  'Yes' 'No']
 ['0.004674348171580075' '1.020025062916693' '-0.15822847016890818'
  '-0.17704431015047875' '-0.06463878920603505' '-0.2188196959070291'
  '-0.21074542005157487' '-0.3075598577650335' 'Male'
  'Non-Hispanic White' 'College graduate or above'
  'Married / living with partner' 'Yes' 'No']
 ['0.34077255767480047' '-0.1415918792927733' '0.8065635746422932'
  '1.2704761127208193' '1.3770375334932605' '0.7735508072544757'
  '-0.08628258461435334' '-0.07577996460848924' 'Female'
  'Non-Hispanic White' 'Some college/AA' 'Married / living with partner'
  'No' 'No']]
```

Centroids clearly differentiate **SES**, **lifestyle**, and **hypertension gradients** across the four clusters.

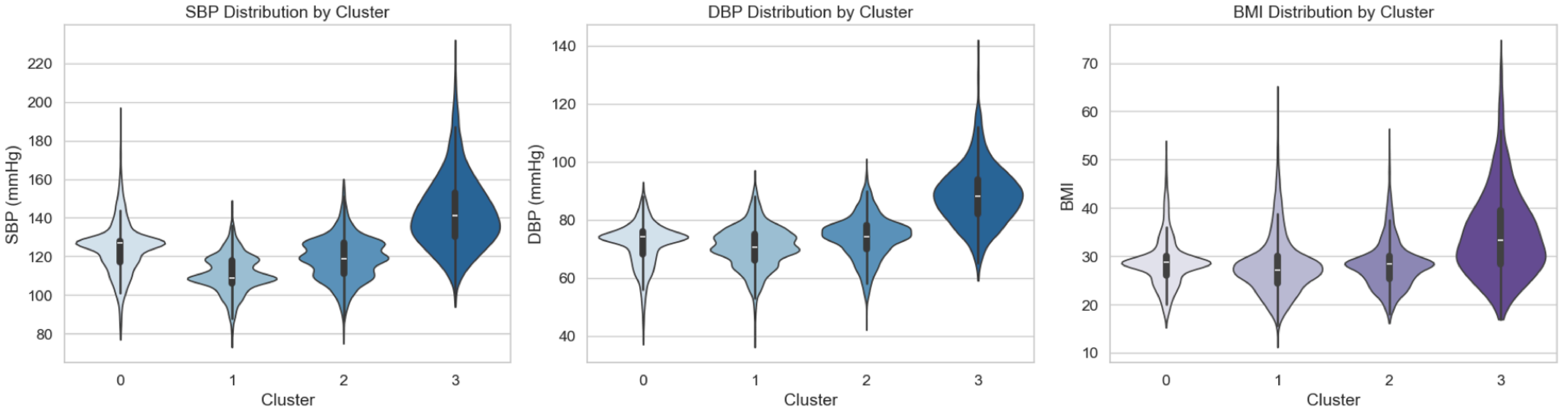
Cluster Profiles

Socioeconomic & Demographic Structure



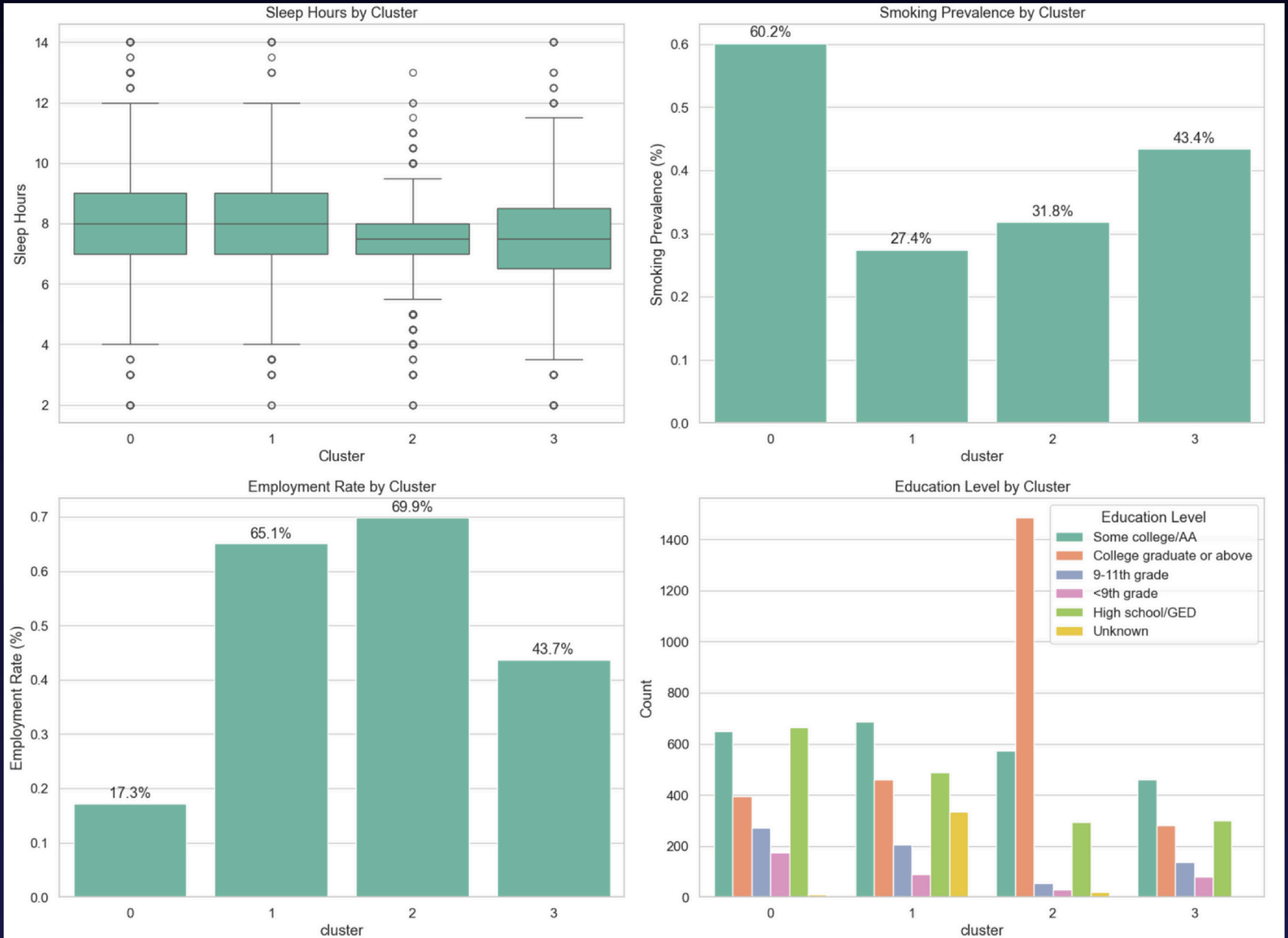
Cluster Profiles

Physiological Health Risk Profile



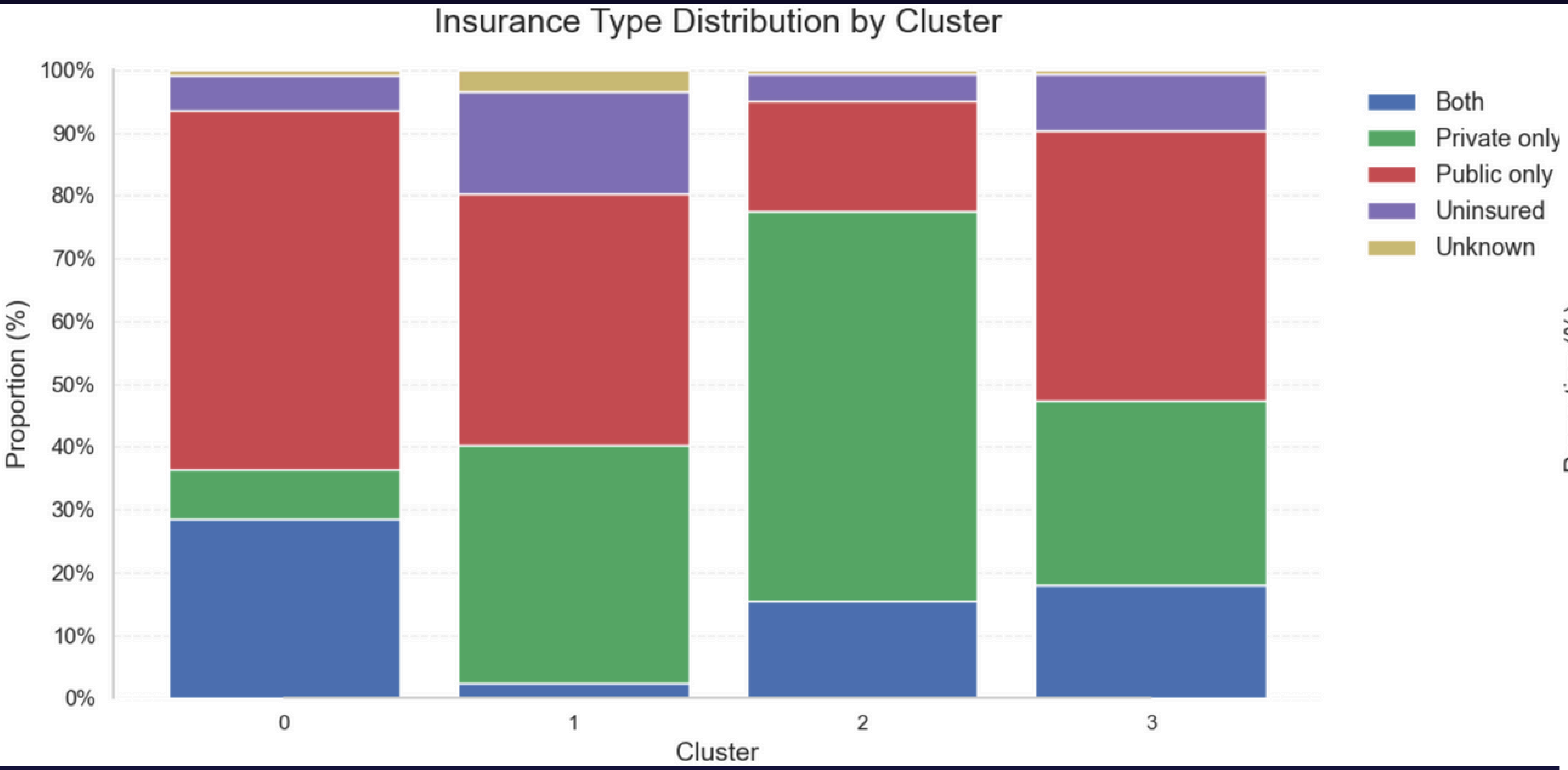
Cluster Profiles

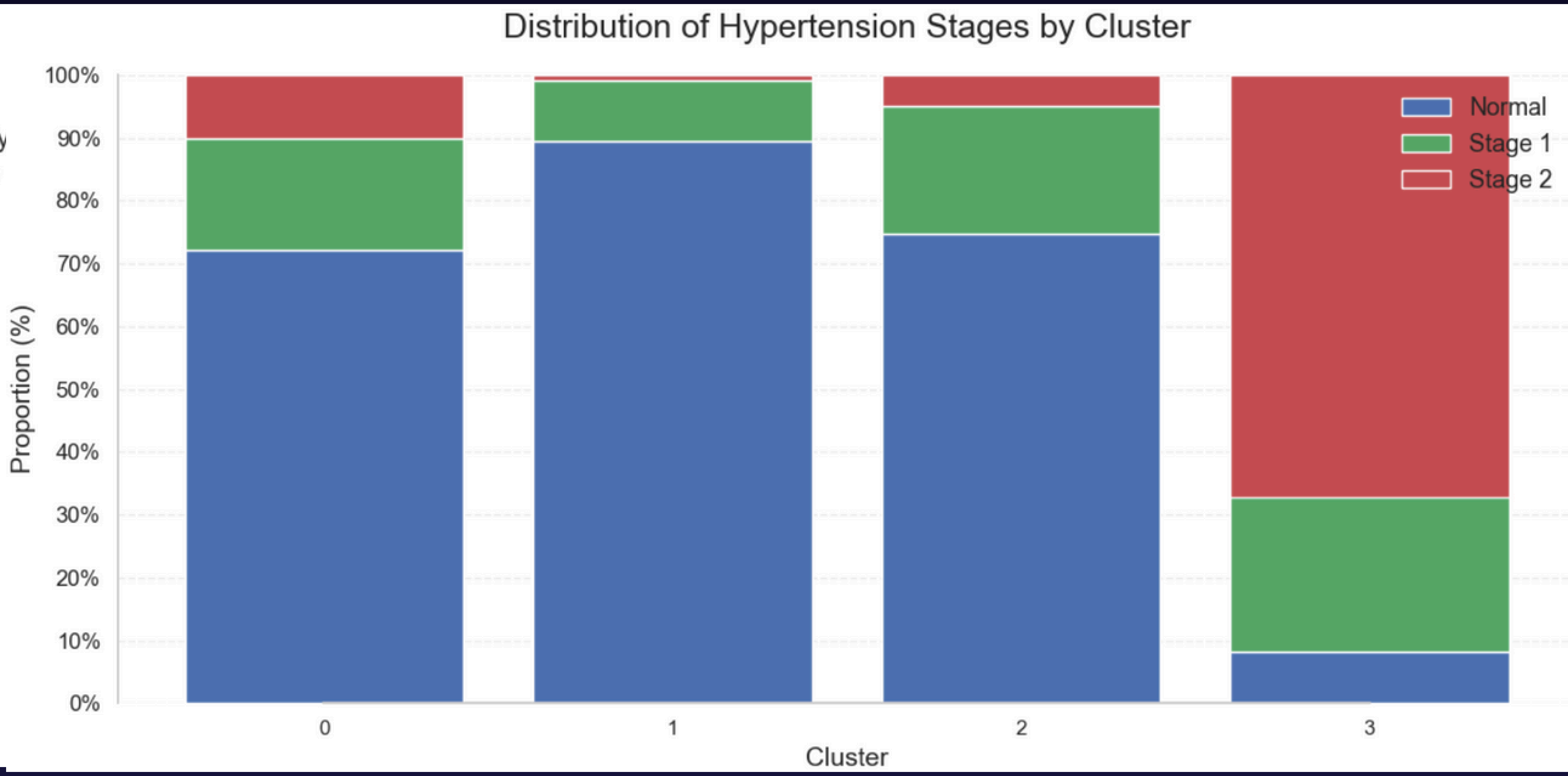
Behavioral & Social Patterns



Cluster Profiles

Insurance Type & Hypertension Stage Distribution





Cluster Interpretation

- ## Cluster 0 Low-Income Lifestyle-Risk Group

Income-to-poverty ratio lowest → clear low-SES pattern
Age is oldest on average (≈ 65–80 yrs)
Smoking prevalence highest (≈ 60%)
Employment rate lowest (≈ 17%)
HTN distribution:
- Normal: 72%
- Stage 1: 18%
- Stage 2: 10%
- Insurance mainly public

- ## Cluster 1 — Young, Healthy, but Uninsured

- Youngest group (≈ 25–40 yrs)
- Highest proportion of Normal BP (≈ 89%)
- Lowest Stage 2 HTN (≈ 1%)
- Employment high (≈ 65%)
- Uninsured rate noticeably higher than other groups
- SES moderately low but stable

- ## Cluster 2 — Well-Resourced, Protected, Low-Risk Group

- Highest income-to-poverty ratio → highest SES
- Education level highest
-Age (≈ 40-65 yrs)
- Employment highest (≈ 70%)
- Private insurance dominates
- HTN distribution:
- Normal: 75%
- Stage 1: 20%
- Stage 2: 5%
- Low BMI → lifestyle relatively healthy

- ## Cluster 3 — Older, High-Risk Hypertension Group

- Age second-highest (≈ 50–70 yrs)
- Stage 2 hypertension extremely high (≈ 67%)
- BMI highest of all clusters
- Smoking high (≈ 43%)
- Insurance type mixed (public + private + uninsured)
- Employment moderate

Conclusion

Key Findings

Insurance Findings

- **Private insurance →** *high-income, college-educated*
- **Public insurance →** *low-income, especially women*
- **Dual**: *older adults on Medicare*
- **Uninsured**: *most heterogeneous → hardest to predict*

Hypertension Findings

- **Stage 1 →** *mainly predicted by BMI*
- **Stage 2 →** *strongest predictor = age*
- **Young adults remain mostly normal BP even when overweight**

Cluster Findings

- **Cluster 0: Low-income, older, smokers** → *moderate HTN*
- **Cluster 1: Young & healthy but uninsured**
- **Cluster 2: High-SES, healthy lifestyle** → mild HTN
- **Cluster 3: High BMI + unstable insurance** → *Severe HTN*

Conclusion

Public Health Implications

1. Low-SES groups accumulate long-term risk

Higher smoking, BMI, older age → higher HTN stages

Implication: *preventive care; lifestyle programs*

2. Publicly insured older adults have unmet needs

Older age + elevated BP + poor lifestyle

Implication: *strengthen follow-up; chronic care within public insurance*

3. Young uninsured adults: healthy now, risky later

Highest uninsured rate

Implication: *affordable insurance options, preventive outreach*

4. Severe HTN + high BMI + unstable insurance

Fragmented care

Implication: *integrated chronic care; insurance continuity*

Conclusion

Limitations & Future Directions

Limitations

- NHANES is cross-sectional
- Self-reported lifestyle data
- High Laboratory Data Missingness

Future Work

- Link NHANES with cost data (e.g., MEPS)
- Use dietary representation learning for richer features

Thank You